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175	Candidate successor manuscript · not peer reviewed · no empirical	
176	planetary validation · no repository modification · no GitHub release,	
177	tag, DOI, or Zenodo action	
178	Repository basis. The public ASTRA repository was audited on 16 August	
179	2026 at this exact main commit:	
	 f8b32ef0af9cb6804f256490b4daafbdba43740e	
180	SPPT/ASTRA v1.0.7 is the current immutable stable reference release;	
181	v1.0.6 is the historical immutable baseline. At that frozen basis, the default	
182	branch was 15 commits ahead of the v1.0.7 tag and contained an explicitly	
183	unpromoted core-integrity M1 source repair plus communications-cover	
184	maintenance. This manuscript is a proposed v1.0.8 successor candidate. It	
185	does not alter the repository, tags, release assets, DOI state, or Zenodo state.	
186	(Jacko T. 2026b , 2026p , 2026o)	
187	Scientific status. This document retains the v1.0.7 scientific classification:	
188	a not-peer-reviewed perspective and mathematical framework with reduced	
189	synthetic demonstrations. It introduces no new astronomical detection, no	
190	dark-matter identification, no proof of planetary topology recovery, no claim	
191	that Newton’s third law fails in a closed fundamental system, and no commer-	
192	cial fuel-cell validation. It separates exact derivations, external experimental	

reports, structural inferences, and proposed tests.

Licensing intent. Original software and schemas proposed by the project remain suitable for MIT licensing; original manuscript text, diagrams, and generated synthetic results remain suitable for CC BY 4.0 to the extent licensable rights exist. Cited publications, scientific facts, repository dependencies, and third-party fonts remain outside that grant.

Abstract

Solar-Planetary Phase-Partition Theory (SPPT) represents a planet as a thermodynamically constrained network of material and energy reservoirs whose physical connectivity may itself be a hidden state. Astronomical State-Topology and Reservoir Analysis (ASTRA) is the inference layer that compares admissible candidate networks against observations, interventions, and simpler baselines. The current stable reference release, v1.0.7, adds stateful edges and operator-aware inference. This v1.0.8 candidate audits the present repository, incorporates its unpromoted source-integrity corrections, and extends the framework to cases in which the hidden source changes the medium that makes the source observable.

The central proposal is the **Endogenous Visibility Principle**: when a source materially alters an envelope, plasma, circumstellar environment, debris field, or detector boundary, the source and the visibility medium must be inferred jointly. A fixed observation operator is then inadequate. Four recent calibration records motivate distinct audits. A dense gas envelope around an early accreting black hole can create a star-like spectrum and bias mass inference. A borderline soft-X-ray trigger becomes an identified engine-driven supernova only through optical, spectroscopic, temporal, and radio information. A reso-

nantly driven early-universe plasma can reorganize and detune itself before the linear dark-photon calculation deposits the assumed energy. Catastrophic destruction in the Neptune system can expose altered parent-body interiors while erasing provenance and chronology. These mechanisms are physically unrelated; they share an inverse-problem structure, not a common hidden substance.

The candidate contributes a stateful visibility equation, source-shell separation, cross-channel rescue, self-detuning seams, catastrophic tomography, and a proposed backreaction-detuning diagnostic. It also reconciles these additions with closure-conditioned reciprocity, mode-resolved active support, sector-complete instruments, observational equivalence, dual-rent promotion, and local-to-global certificates. A proposed benchmark suite requires hidden generators, transducers, observation channels, and archive operators to compete under conserved ledgers, out-of-set controls, and held-out tests.

This is a methodological successor candidate, not an empirical planetary validation, dark-matter detection, proof of a universal arrow of time, or demonstration that the calibration cases share one physical law. Its governing standard is narrower: identify the carrier, model the stateful medium, declare what the instrument cannot distinguish, change the observation operator, and preserve the failed prediction.

Executive summary

What the repository currently contains

The current repository has a deliberately stratified architecture. SPPT/ASTRA v1.0.7 is the immutable stable core reference release, while v1.0.6 remains

the historical immutable baseline. *Earth Is the Instrument* v0.3.0 is a separately versioned foundational supplement. The Sector-Complete Instrument is a public namespaced alpha module. Active Support, the Bridge Protocol, Cosmic Visibility, and Coherence-Cell Exploration are public but unpromoted drafts or prototypes. The repository documentation explicitly states that those resources do not automatically enter the v1.0.7 claim-admission matrix or inherit its release verification. (Jacko T. 2026b)

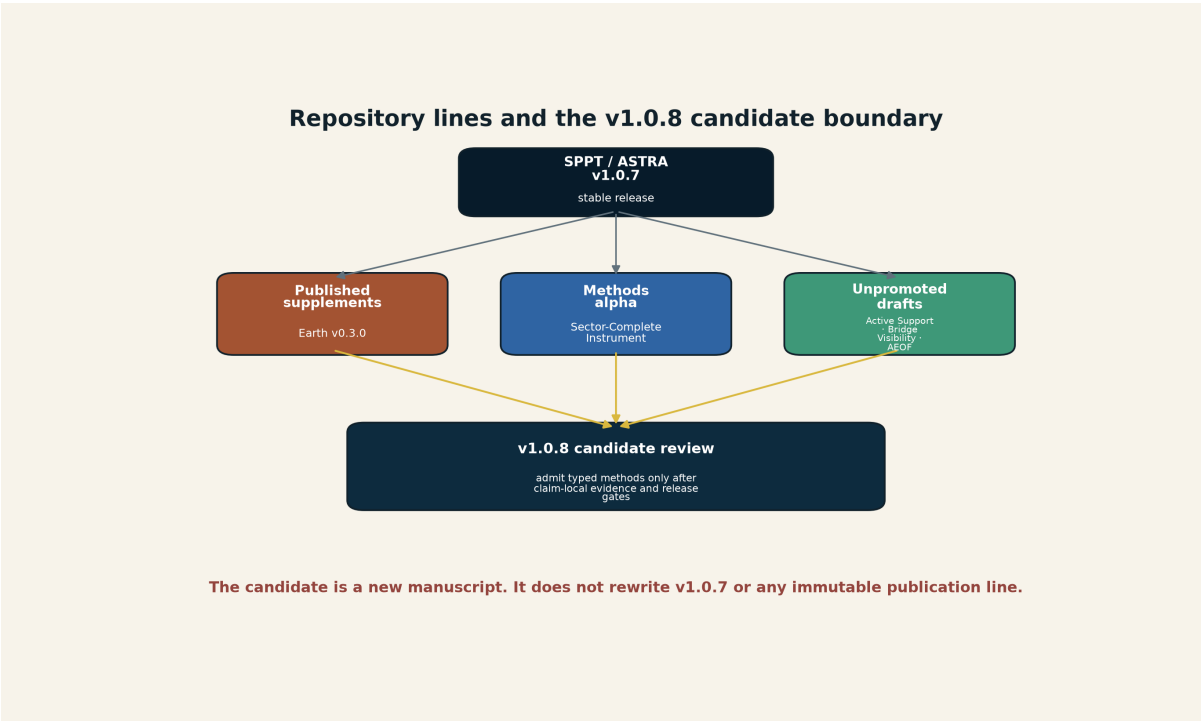


Figure 1: [MODEL] Current repository architecture and the proposed v1.0.8 candidate-admission boundary. The arrows denote review and candidate admission, not scientific endorsement or physical causation. Creator: ASTRA / Jacko T. Source: original vector model for this document. License: CC BY 4.0.

The appropriate next step is not to concatenate every document. It is to identify the common typed objects that survived red-team review and admit them with precise scope while preserving domain-specific evidence boundaries.

252 **The retained central object: a stateful edge**

253 The stable core, inherited from v1.0.6 and integrated in v1.0.7, makes the
254 graph part of the physical state and makes edge state explicit. The v1.0.8
255 candidate retains that stateful-edge representation:

$$\mathcal{P}_{\text{phys}}(t) = (\mathcal{G}(t), x(t), \theta, u(t), b_E(t)), \quad b_E(t) = \{b_e(t) : e \in E(t)\}.$$

256 Here x contains continuous node or field variables, θ contains constitutive
257 parameters, u is forcing or control, and b_e is the state of edge e . The edge
258 state is not an unrestricted metaphor. It is admitted only when measurable
259 storage, hysteresis, relaxation, selectivity, geometry, damage, composition, or
260 history improves prediction or intervention discrimination.

261 The constitutive and evolution laws are

$$J_e = G_e(\Delta X_e, b_e, u, \mathcal{A}_e), \quad \dot{b}_e = F_e(b_e, x, u, J_e),$$

262 where $\mathcal{A}_e$ is a mode-resolved active-support descriptor. A topology change
263 means an edge or node appears, disappears, connects, or disconnects. A
264 change in b_e , its weight, directionality, or active support does **not** automat-
265 ically mean topology changed. That separation is essential.

266 **The retained closure rule**

267 An effectively nonreciprocal interaction in a reduced subsystem is not permis-
268 sion to discard conservation. Let P be the observed particle subsystem and
269 E the environment. Then

$$\frac{d\mathbf{P}_P}{dt} = \mathbf{F}_{P \leftarrow P}^{\text{eff}} + \mathbf{J}_{E \rightarrow P} + \mathbf{F}_{\text{boundary}}.$$

If the reduced pair forces do not cancel, the residual must be assigned to fluid, field, electrode, substrate, controller, or another declared environment channel. The candidate therefore adds **closure-conditioned reciprocity**:

A directed effective interaction is admissible only when the enlarged momentum, energy, charge, species, and entropy ledgers specify what entered, left, or was dissipated.

The retained observation rule

The instrument cannot be represented by one generic arrow. The v1.0.8 candidate composes distinct operators:

$$H \xrightarrow{\Phi_t} X \xrightarrow{\mathcal{S}_{\Gamma, u}} Y \xrightarrow{\mathcal{T}} Z \xrightarrow{\mathcal{V}} D \xrightarrow{\mathcal{C}} C.$$

H is a candidate history, Φ_t the physical evolution, $\mathcal{S}$ a selective or stateful seam, $\mathcal{T}$ a carrier transformation, $\mathcal{V}$ visibility, sampling, and detector response, and $\mathcal{C}$ a bounded certificate. None of these operators may be silently substituted for a physical matter or energy edge.

A protocol π certifies only the observational quotient

$$K_i \sim_{\pi} K_j \iff p(D \mid K_i, \pi) = p(D \mid K_j, \pi).$$

“Sector complete” means that the plausible sectors in the declared candidate family have been enumerated and measured or bounded. It does not mean

every imaginable sector has been measured.

What the two new studies add

The nonreciprocal-colloid experiment adds a concrete physical reason to track directed effective edges, environment-mediated momentum exchange, and dynamic graph renewal. The important phenomenon is not merely movement. Nonreciprocity prevents ordinary coarsening from completing; the cluster topology remains dynamically reconfigurable.

The Cu₃Pt experiment adds a concrete physical reason to split edge state into fast reversible control and slow irreversible history. Compression changes surface electronic structure; operation changes surface composition through selective Cu loss. The edge law is therefore path dependent even if bulk material labels remain unchanged.

Together, the studies support a new candidate rule:

When operation changes the edge that performs the operation, model the edge as a self-rewriting subsystem.

What the accelerating-wave paper adds - and does not add

The linked August 2026 article is not reporting a newly published 2026 theory. It is a new popular account of Koivurova, Robson, and Ornigotti's 2023 Optica paper. The paper begins from the one-way transport relation

$$(\partial_t + c \partial_x) f = 0$$

and, for prescribed $c = c(t)$, derives

$$\partial_t^2 f = c(t)^2 \partial_x^2 f - \dot{c}(t) \partial_x f.$$

The formalism is relevant to ASTRA because it makes the **control history of** 306
a medium part of the wave problem. It also forces a reference-frame audit: 307
frequency, wavelength, momentum, and the paper’s intrinsic-time coordinate 308
are not interchangeable observables. The authors explicitly keep global en- 309
ergy conservation by including the pump that drives a time-varying medium. 310
(Koivurova, Robson, and Ornigotti 2023) 311

The stronger claim - that the positive-time branch supplies a universal 312
microscopic arrow of time - remains theoretical and is not established by the 313
existing experiments on temporal interfaces. Time-reflection experiments 314
have demonstrated frequency translation and waveform reversal under 315
externally switched media while laboratory time continues forward. They do 316
not show time itself reversing, but neither do they by themselves prove that 317
the accelerating-wave branch choice is a fundamental law of microscopic 318
time. (Galiffi et al. 2022; Moussa et al. 2023) 319

The correct ASTRA placement is therefore: 320

Published theoretical calibration; useful temporal-interface 321
audit; universal arrow-of-time interpretation open. 322

The new visibility feedback loop 323

The v1.0.7 operator stack is extended from a fixed visibility map to a stateful 324
one: 325

$$D(t) = \mathcal{O}_\pi \left[\mathcal{V}_{b(t)}(S(t)) \right] + \epsilon(t), \quad \dot{b} = F(b, S, \mathcal{V}_b[S], u).$$

326 S is the hidden source, b the state of the envelope, plasma, circumstellar ma-
 327 terial, debris field, or detector boundary, and $\mathcal{V}_b$ the state-dependent trans-
 328 duction and propagation operator. A fixed-operator approximation is justified
 329 only when the source-induced change in b is negligible over the inference win-
 330 dow.

331 This yields four new audit objects:

- 332 • **source-shell separation** - distinguish the engine from the atmosphere or
 333 medium that shapes its spectrum;
- 334 • **cross-channel rescue** - record when an ambiguous event becomes identi-
 335 fiable only through an orthogonal messenger or trigger;
- 336 • **self-detuning seam** - test whether the signal changes the resonance or
 337 detector before the claimed signal accumulates;
- 338 • **catastrophic tomography** - model destruction as an operator that may
 339 expose an interior while destroying provenance and chronology.

340 **Role-based coherence cell**

341 The manuscript was reviewed through a role-based coherence cell rather than
 342 by a claimed external committee. Each role was allowed to certify only its own
 343 link in the chain; no role was permitted to validate the complete framework.

	Primary responsibility in this
Working perspective	candidate
Planetary physicist and geochemist	reservoir legality, phase interpretation, planetary scope

Working perspective	Primary responsibility in this candidate
Transient and high-redshift astrophysicist	source-shell separation, trigger selection, multiwavelength alternatives
Plasma physicist	resonance, nonlinear saturation, collision and scale limitations
Planetary dynamicist and spectroscopist	Triton-capture histories, mineral interpretation, archive remapping
Condensed-matter and active-matter physicist	nonreciprocal reduced dynamics and environmental closure
Electrochemist and materials scientist	fast strain control, slow surface evolution, full-cell translation limits
Applied mathematician and control theorist	observability, equivalence classes, local-to-global scope
Statistician and experimental designer	identifiability, nuisance structure, held-out interventions
Scientific editor, accessibility reviewer, and visual-information designer	paragraph structure, line spacing, figure legibility, provenance, and document QA
Adversarial red team	ordinary alternatives, overclaim detection, demotion conditions, failure memory

This matrix describes audit responsibilities, not external endorsement, peer review, institutional affiliation, or authorship.

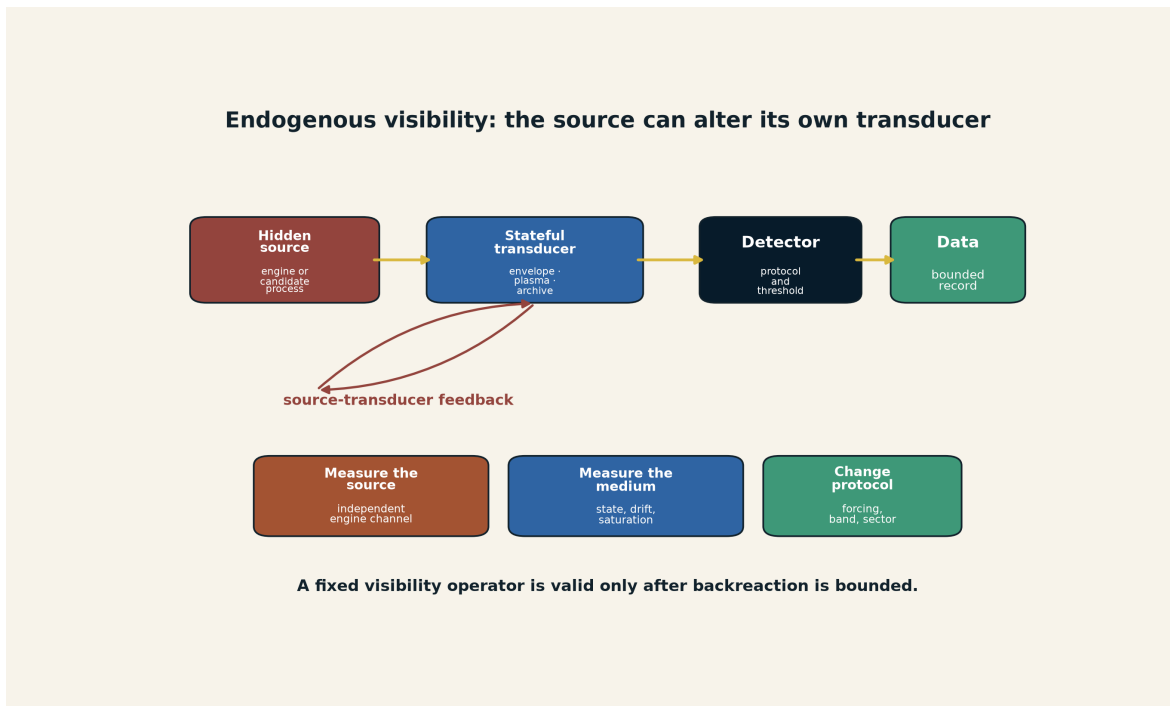


Figure 2: [MODEL] Endogenous visibility. A hidden source drives a transducer whose changing state feeds back on the conversion into data. Independent measurements of the transducer state and altered observation protocols are required to separate source from medium. Creator: ASTRA / Jacko T. Source: original vector model for this candidate. License: CC BY 4.0.

What remains unchanged

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The v1.0.6 core claims remain the scientific anchor: network inventory
accounting, local entropy-production conditions, the exact periodic-trap
solution, the weak-cut spectral bound, the corrected derivative identity,
the heterogeneous-nucleation wetting factor, static deep-conductance non-
identifiability, and the bounded synthetic benchmark results. The candidate
does not upgrade any of those from conditional mathematics or synthetic
evidence to planetary validation. (Jacko T. 2026h, 2026n, 2026m)

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The candidate also preserves every important negative boundary:

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- no evidence that the new colloidal effect is a new fundamental force;
- no evidence that the catalyst is commercially durable or full-cell validated;
- no evidence that active-support notation is a new universal constitutive law;
- no evidence that sector conversion identifies dark matter;
- no evidence that cosmic visibility papers detect gravitons or dark-sector identity;
- no evidence that mathematical local-to-global analogies imply a simulated or holographic universe;
- no experimental proof that the accelerating-wave equation establishes a universal microscopic arrow of time;
- no claim that one physical mechanism unifies planets, cells, catalysts, active matter, monuments, and arithmetic.

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Candidate status verdict

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This document is suitable as a **full scientific integration candidate**. It is
not suitable for immediate release as v1.0.8 without a frozen source tree, ex-
act claim-source coverage, code integration, test extension, two independent

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371 builds, full accessibility review, release-identity binding, and natural main/tag
372 workflow passes. The proposed release path is given in Part VII.

373 **Epistemic and implementation vocabulary**

374 Two vocabularies are retained because they answer different questions.

375 **Scientific status** describes what the world-facing evidence supports:

Label	Meaning
Established	Repeatedly observed or strongly anchored by direct measurement and mature theory within a stated domain.
Strong inference	Best explanation of convergent evidence; details remain revisable.
Plausible	Physically coherent and partially supported.
Open	Not excluded; current evidence does not materially favor it.
Constrained	Possible only in narrowed forms because expected evidence is absent or contradictory.
Unsupported	No positive evidence currently requires the claim.

376 **Evidence class** describes how a project statement was supported:

Class	Meaning
source_asserted	Present in an identified external or supplied record; not independently reproduced.
hand_checked	Algebra, dimensions, signs, or logical scope checked directly.
independently_reproduced	Recomputed from declared inputs without relying on a saved transcript.
mechanically_replayed	Deterministic project code or certificate replayed under a declared runtime.
externally_published	The external record has a stable publication identity.
structural_inference	A cross-domain method comparison that preserves different physical laws.
proposed_only	New notation, test, or architecture awaiting data or proof.
deferred	Blocked by source, rights, units, implementation, or control gaps.
rejected	A tempting interpretation contradicted by the present evidence or scope.

No evidence class by itself establishes truth. A mechanically replayed synthetic result may be exact and still fail to describe a planet. An externally

published paper may be real and still not entail ASTRA's structural interpretation.

Part I - Repository baseline and retained core

1. Current repository state

1.1 Stable release and default-branch separation

SPPT/ASTRA v1.0.7 is the current stable core reference edition and GitHub Latest release. Its formal classification remains a not-peer-reviewed perspective and mathematical framework with reduced synthetic demonstrations and no empirical planetary validation. The release has its own tag, source archive, claim matrix, source inventory, generated PDF/HTML editions, checksums, and identity record. (Jacko T. 2026p, 2026k, 2026j)

The repository default branch is not identical to the immutable v1.0.7 release tree. At the audited commit it is 15 commits ahead of the tag. Those commits include an unpromoted core-integrity M1 repair and a communications-cover redesign. The README states explicitly that the M1 source differs from the immutable reading assets and is neither v1.0.8 nor a published erratum. (Jacko T. 2026b, 2026o)

This separation is scientifically important. A public repository can simultaneously contain:

- a frozen citation target;
- a later source repair that has not been promoted;
- supporting evidence overlays;

- communications-only changes; 401
- separately versioned supplements and research previews. 402

A current file is not automatically an admitted claim. A passing repository 403
gate is not external scientific validation. The release identity and evidence 404
status must travel with the claim. 405

1.2 Core-integrity M1 corrections that this candidate 406 adopts 407

The M1 repair identified three consequential wording defects in the v1.0.7 408
source and corrected them without rewriting the immutable release assets. 409

First, the weak-cut statement is capacity-weighted. For a connected positive- 410
conductance graph with strictly positive node capacities and a nonempty 411
proper cut, weak conductance relative to the aggregate capacities on both 412
sides gives a small Rayleigh quotient and therefore a slow-relaxation upper 413
bound. Low capacity alone does not imply slow relaxation. 414

Second, the earlier asymptotic logarithmic-slope criterion for dynamic arrest 415
was not sufficient. An unbounded process such as $\ell(t) = \log(1 + t)$ has log- 416
arithmic slope tending to zero, whereas a bounded oscillatory process can 417
fail that slope test. The repaired operational record therefore requires a pre- 418
registered boundedness or statistical-stationarity assessment over a declared 419
ensemble and window, plus persistent turnover above a positive threshold. It 420
remains a proposed observational classification, not a universal physical law. 421

Third, the earlier claim-source-completeness sentence was too strong. The 422
maintenance record reports structural path support for 55 claims and exact lo- 423
cators for 21. External entailment was not comprehensively reverified; source 424

hashes and retrieval dates are incomplete; and claim-local replay commands, runtimes, and run identifiers remain incomplete for some computational labels. The correct description is **maintenance evidence with explicit coverage limits**, not sentence-level closure. (Jacko T. 2026o)

1.3 Supplemental and research lines

Earth Is the Instrument v0.3.0 remains a separately versioned foundational working paper. It develops boundary-state promotion, typed physical/control/observation/archive/certificate relations, FOG audits, seam information, dual rent, residue fields, archive-veto strength, local-to-global certificates, and comparative origin ledgers. The central proposition - Earth as reactor, archive, censor, and instrument - remains supplemental to the planetary core rather than an automatically admitted physical law. (Jacko T. 2026a, 2026f)

The Sector-Complete Instrument alpha remains a namespaced research preview. Active Support, the SPPT Bridge Protocol, Cosmic Visibility, and Coherence-Cell Exploration remain public but unpromoted methods drafts or prototypes. Their useful contributions are typed records, observational quotients, visibility operators, intervention design, and analogy-to-falsifier discipline. None inherits the stable release identity merely by residing on main. (Jacko T. 2026c, 2026g, 2026l, 2026e, 2026d)

1.4 Why v1.0.8 should be a successor rather than a silent patch

The new astrophysical and plasma cases do not merely add examples to v1.0.7. They expose a missing feedback loop: the source can alter the medium that

converts it into an observation. The existing operator-aware visibility frame- 449
work treats transduction and sampling explicitly, but its simplest expression 450
can still be read as a fixed operator acting on a source. The new candidate 451
makes the transducer state dynamical and source-coupled. 452

The admission strategy is therefore: 453

		v1.0.8 candidate disposition
Layer	Content	
Stable physical core	v1.0.7 conservation, thermodynamics, topology, stateful edges	retain with M1 wording corrections
Visibility dynamics	source-coupled transducer state and backreaction	propose as new ASTRA method
Cross-channel inference	trigger-aware rescue and instrument-conditioned categories	admit as bounded observation method
Catastrophic archives	destruction, transport, and reaccretion as information operators	admit as archive method
External calibration	black-hole envelopes, X-ray transients, plasma saturation, Neptune debris	cite as domain-specific evidence only

Layer	Content	v1.0.8 candidate disposition
Supplemental origins line	geological memory, human archive, comparative origins	keep separately versioned
Mathematical calibration	local-to-global and prime-reduction lessons	retain as certificate analogies, not cosmological evidence

2. Retained SPPT physical architecture

2.1 Nodes, edges, and inventories

A node $v \in V$ represents a spatially or thermodynamically distinguishable reservoir with assignable state and inventory at the model's resolution. An edge $e = (a \rightarrow b, p)$ states that process p can transfer matter, energy, charge, momentum, or a declared species from a to b .

Let $M \in \mathbb{R}^{n \times s}$ hold node inventories, $B \in \mathbb{R}^{n \times m}$ the directed incidence matrix, $J \in \mathbb{R}^{m \times s}$ edge fluxes, $R \in \mathbb{R}^{n \times r}$ local reaction rates, $N \in \mathbb{R}^{s \times r}$ stoichiometry, and $S, E \in \mathbb{R}^{n \times s}$ external supply and removal. The retained species balance is

$$\dot{M} = BJ + RN^T + S - E.$$

If w encodes a conserved elemental or charge combination with $N^T w = 0$, then

$$I_w = \mathbf{1}^\top M w, \quad \dot{I}_w = \mathbf{1}^\top (S - E) w.$$

The candidate does not replace this matrix balance. Stateful edges modify how J is calculated and how the model records omitted environment exchange. They do not create a new source term by rhetoric.

2.2 Thermodynamic closure

For conjugate edge forces X_e and fluxes f_e , the local entropy-production condition remains

$$\dot{S}_{i,e} = f_e^\top X_e \geq 0$$

on the declared domain, with the appropriate sign and temperature conventions. Near equilibrium a linear closure $f_e = L_e X_e$ is admissible when the dissipative symmetric part of L_e is positive semidefinite. Far from equilibrium, the application must supply its own constitutive law and entropy or free-energy accounting.

The new nonreciprocity layer does not repeal this condition. It makes the environment and coarse-graining boundary explicit so that apparently non-conservative pair dynamics are not confused with a closed thermodynamic description.

2.3 Memory, traps, and bottlenecks

The v1.0.6 periodic trap remains the simplest edge-memory calibration:

$$\dot{M} = c_0 + c_1 \cos(\omega t) - \frac{M}{\tau}.$$

Its steady periodic solution, phase lag, and loop integrals show how a release time τ produces memory under periodic forcing. The raw loop magnitude and release-normalized loop magnitude have different dependence on $\omega\tau$; those quantities must not be conflated.

The weak-cut spectral result remains the simplest topological bottleneck calibration. For a connected positive-conductance graph with strictly positive node capacities and a nonempty proper cut, weak cut conductance relative to the aggregate capacities on both sides yields a small Rayleigh quotient and therefore a slow-relaxation upper bound. Low capacity alone does not imply slow relaxation. A slow mode can therefore arise from topology even when local constitutive laws are ordinary.

The v1.0.8 candidate adds another possibility: a slow mode can arise because an edge state evolves slowly, because its active support turns on intermittently, or because a directed effective coupling maintains dynamic reorganization. Those mechanisms must be separated by intervention.

2.4 Static non-identifiability remains the baseline warning

In the retained two-reservoir closure, the same static surface equilibrium can coexist with different deep conductance and hidden deep temperature. The synthetic supplement extends that lesson: four connected three-node graph families can share one static surface equilibrium while holding different interior states, and multi-frequency or held-out forcing supplies additional discrimination. ([Jacko T. 2026i](#))

This is the baseline against which the new methods should be judged. An edge-
state variable is useful only if it explains data that fixed topology and fixed
edge parameters cannot explain, and if the added variable remains identifiable
under an improved observation protocol.

3. Candidate integration axioms

The five v1.0.6 axioms are retained. The candidate adds four subordinate
axioms. They do not override conservation or thermodynamics.

Axiom A6 - Edge-state explicitness. If the current and future flux across
edge e depend on an interfacial history not contained in the adjacent node
states, represent that history by a declared edge state b_e or show that a re-
duced memory kernel is sufficient.

Axiom A7 - Closure-conditioned reciprocity. If a reduced interaction is
nonreciprocal, identify the external drive, mediator, substrate, controller,
fluid, field, or boundary that closes momentum, energy, charge, and entropy
accounting in the enlarged system.

Axiom A8 - Observation-sector explicitness. A null or positive result con-
strains only the sectors, carriers, resolutions, and nuisance model contained
in the observation operator. The unresolved generator quotient must be re-
ported.

Axiom A9 - Operator promotion by rent. An active-support, visibility, sec-
tor, or edge-state variable is retained only if it changes reachable outcomes,
improves generator discrimination, or supplies an exact closure certificate
under predeclared testing. Otherwise it remains bookkeeping or is removed.

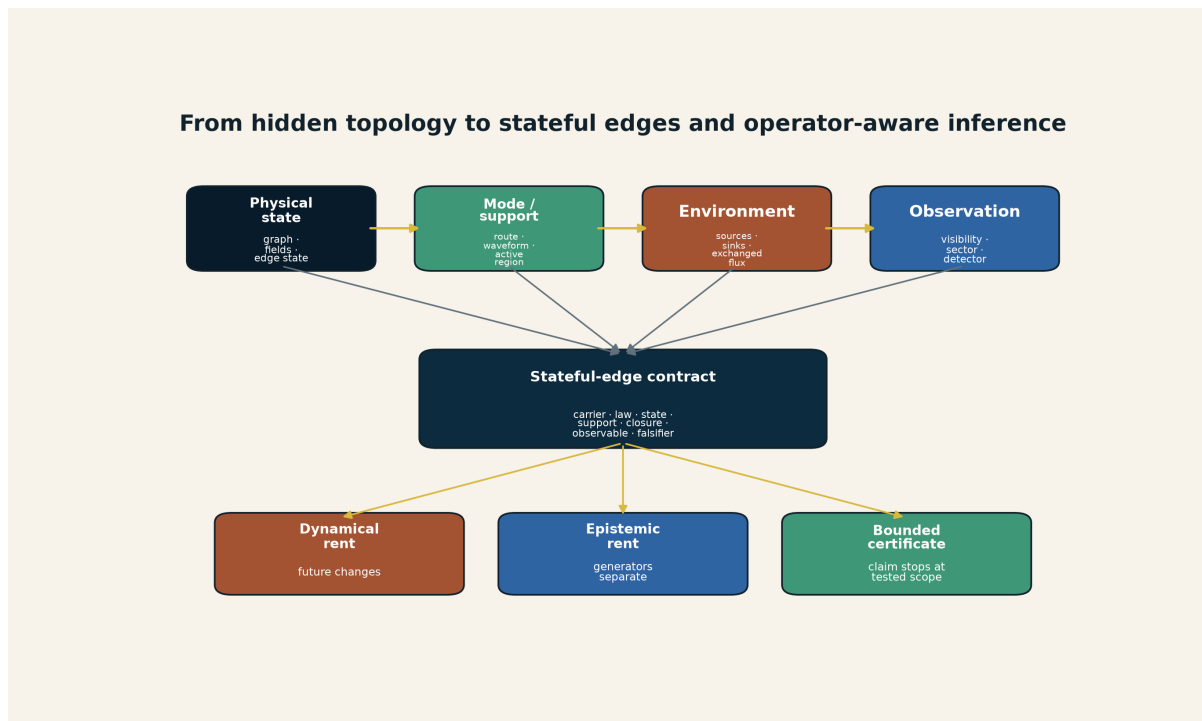


Figure 3: [MODEL] Expanded state architecture. Physical state, mode/support, environment, and observation remain typed and feed a stateful-edge contract. The lower boxes separate dynamical rent, epistemic rent, and global certificate scope. Creator: ASTRA / Jacko T. Source: original vector model. License: CC BY 4.0.

Part II - Stateful edges and closure-conditioned nonreciprocity

4. The stateful-edge representation

4.1 Physical state versus inference state

The candidate separates the world model from the inference record:

$$\mathcal{M} = (\mathcal{P}_{\text{phys}}, \Pi_{\text{mode}}, \Pi_{\text{obs}}, C),$$

with

$$\mathcal{P}_{\text{phys}} = (\mathcal{G}, x, \theta, b_E, u),$$

$$\Pi_{\text{mode}} = (\mu, \mathcal{A}, \mathcal{R}_{\text{recip}}, \mathcal{L}_{\text{env}}),$$

$$\Pi_{\text{obs}} = (\mathcal{V}, \mathcal{S}, \mathcal{O}, \mathcal{N}).$$

μ is operating mode, $\mathcal{A}$ active support, $\mathcal{R}_{\text{recip}}$ the reciprocity record, $\mathcal{L}_{\text{env}}$ the environment exchange ledger, $\mathcal{V}$ the visibility/sampling operator, $\mathcal{S}$ the sector set, $\mathcal{O}$ the detector basis, and $\mathcal{N}$ nuisance/calibration state.

This partition prevents a common category error. A detector channel may improve inference without becoming a physical transport edge. A support mask may identify where a reaction occurs without constituting a new mater-

540 ial reservoir. A certificate may validate an equation without proving that the
 541 equation describes a planet.

542 **4.2 The edge contract**

543 A candidate physical edge is represented by the record

$$e = (a, b, q, G_e, \mathcal{D}_e, U_e, b_e, \mu_e, \mathcal{A}_e, \mathcal{R}_e, \mathcal{L}_e, \mathcal{O}_e, F_e),$$

544 where:

Field	Requirement
a, b	tail and head reservoirs or boundary ports
q	transported quantity or transformed species
G_e	constitutive law
$\mathcal{D}_e$	domain, boundary conditions, and quantifiers
U_e	units, sign convention, and reference state
b_e	measurable edge/interface state
μ_e	operating direction or mode
$\mathcal{A}_e$	active support and normalization
$\mathcal{R}_e$	reciprocal, nonreciprocal, odd, or unresolved reduced coupling class
$\mathcal{L}_e$	source/sink and environment-exchange ledger

Field	Requirement
$\mathcal{O}_e$	observation and calibration channels
F_e	falsifier or demotion test

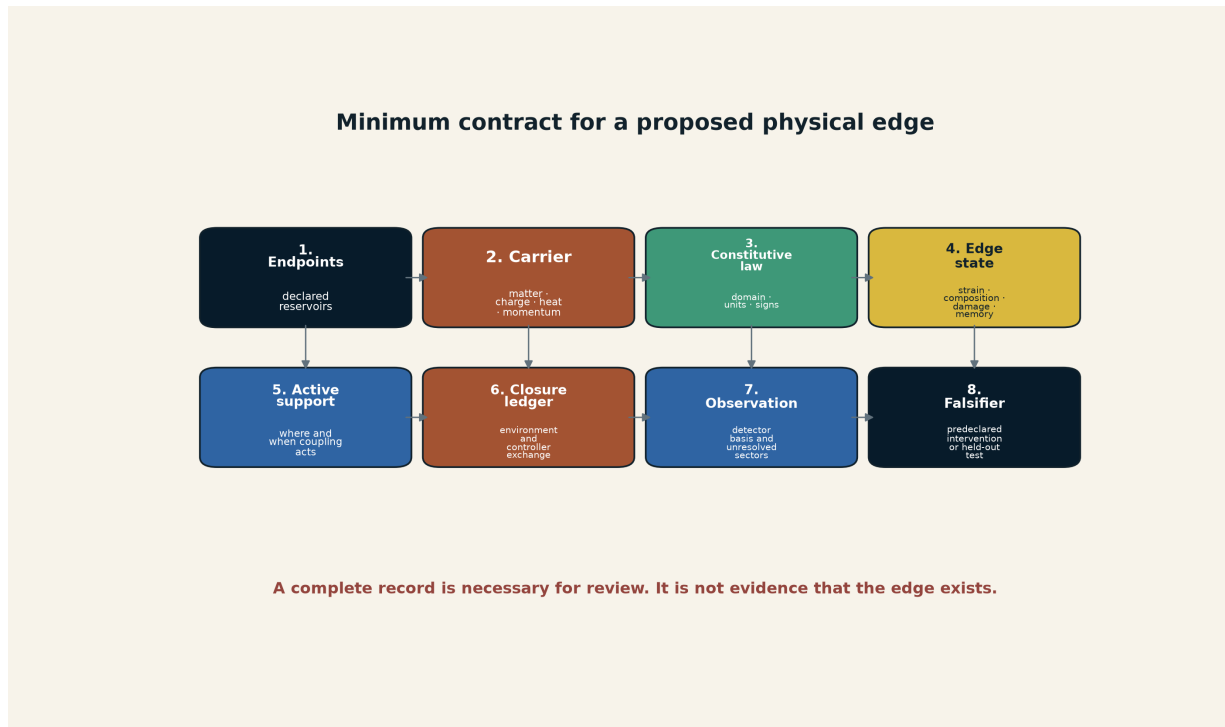


Figure 4: [MODEL] The v1.0.8 candidate edge contract. A typed record is the minimum information required before an edge is inferred or promoted. It is not evidence that the edge exists. Creator: ASTRA / Jacko T. Source: original vector model. License: CC BY 4.0.

4.3 Edge-state promotion

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Begin with a memoryless law

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$$J_e = G_e(x_a, x_b; \theta_e).$$

Promote an edge state when the reduced law is not sufficient:

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$$J_e = G_e(x_a, x_b, b_e, u; \theta_e), \quad \dot{b}_e = F_e(b_e, x_a, x_b, J_e, u).$$

The operational test is:

Hold adjacent bulk states and nominal forcing approximately fixed.
Change the edge history or edge intervention. If the future flux or output distribution changes in a reproducible, held-out way, the edge contains predictive state omitted by the memoryless model.

The edge earns independent state only when at least one of the following is measurable: stored stress, charge, heat, matter, or chemical potential; a relaxation time; hysteresis; state-dependent permeability; evolving composition; damage or healing; active geometry; defect density; wetting or adsorption state; a controller memory; or a prospective intervention that changes output while bulk variables remain approximately fixed.

4.4 Stateful weight change is not topology change

The project must distinguish three levels:

1. **fixed topology, fixed edge law:** only node states and inputs vary;
2. **fixed topology, stateful edge law:** the edge persists but its conductance, selectivity, support, or directionality evolves;
3. **topology change:** an edge or node appears, disappears, connects, disconnects, merges, or splits under a declared guard and reset rule.

Many apparent topology claims are actually level 2. Catalyst strain, fault permeability, receptor populations, and colloidal coupling strength may change dramatically while the named reservoirs remain connected. Conversely, a percolation threshold or phase separation can create a genuine level-3 change.

5. Closure-conditioned reciprocity

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5.1 What nonreciprocity means in a reduced model

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For an effective pair law, Newtonian reciprocity would require

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$$\mathbf{F}_{ij}^{\text{eff}} = -\mathbf{F}_{ji}^{\text{eff}}.$$

Define the pair residual

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$$\mathbf{N}_{ij} = \mathbf{F}_{ij}^{\text{eff}} + \mathbf{F}_{ji}^{\text{eff}}.$$

A nonzero $\mathbf{N}_{ij}$ means the pair subsystem is not closed under that effective description. It does not by itself show that momentum is created. In a driven colloid, the residual may be supplied by the electric field, surrounding fluid, electrode boundary, viscous drag, or an eliminated mediator.

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For the observed particle set P and environment E ,

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$$\frac{d}{dt}(\mathbf{P}_P + \mathbf{P}_E) = \mathbf{F}_{\text{ext}},$$

while

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$$\frac{d\mathbf{P}_P}{dt} = \sum_{i < j} \mathbf{N}_{ij} + \mathbf{J}_{E \rightarrow P} + \mathbf{F}_{\text{boundary}}.$$

The exact partition depends on the coarse-graining. The identity is bookkeeping: every reduced residual must be assigned to a modeled or explicitly omitted exchange channel.

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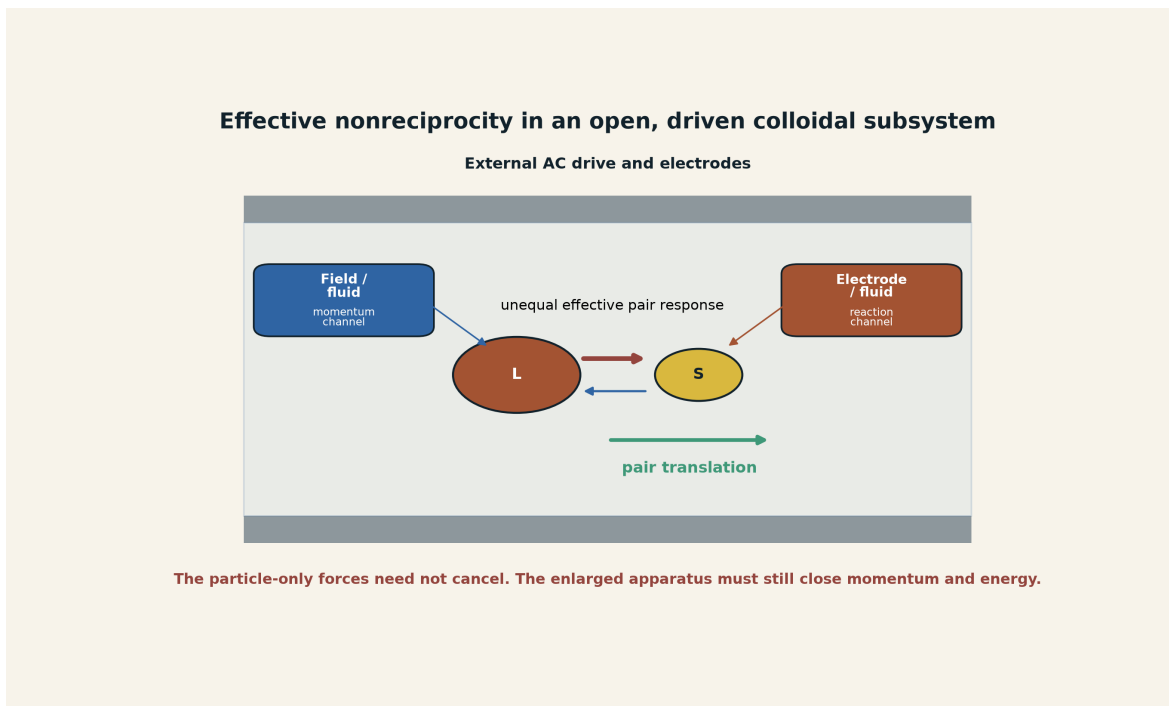


Figure 5: [MODEL] Effective nonreciprocity in an open, driven colloidal subsystem. Unequal particle-level attraction can generate pair translation while momentum and energy are exchanged with the fluid, field, and electrodes. This schematic does not reproduce the experiment’s flow field and does not claim a fundamental violation of momentum conservation. Creator: ASTRA / Jacko T. Source: original vector model informed by Hara et al. License: CC BY 4.0.

5.2 Coupling-matrix decomposition

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For a reduced linearized interaction matrix K , write

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$$K = K^S + K^A, \quad K^S = \frac{K + K^\top}{2}, \quad K^A = \frac{K - K^\top}{2}.$$

K^S is the reciprocal symmetric component in the chosen variables; K^A is the antisymmetric directed component. A model-dependent nonreciprocity index may be recorded as

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$$\eta_{\text{nr}} = \frac{\|K^A\|_F}{\|K^S\|_F + \epsilon},$$

with declared normalization ϵ . This is not a universal physical constant. It depends on coordinates, coarse-graining, mobility, and the choice of interaction variables. It is useful only as a finite model diagnostic.

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5.3 Exact two-particle drift identity

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A minimal one-dimensional reduced model clarifies how asymmetric attraction creates a translating pair. Let $x_L < x_S$, $r = x_S - x_L > 0$, and

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$$\dot{x}_L = a_{LS}r, \quad \dot{x}_S = -a_{SL}r.$$

Then

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$$\dot{r} = -(a_{LS} + a_{SL})r,$$

and the pair center $x_c = (x_L + x_S)/2$ obeys

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$$\dot{x}_c = \frac{a_{LS} - a_{SL}}{2} r.$$

With a short-range repulsion or excluded-volume constraint fixing a finite separation r_* , the pair translates at

$$v_{\text{pair}} = \frac{a_{LS} - a_{SL}}{2} r_*.$$

This is an exact consequence of the toy equations. It is not an independent fit to the experiment and does not determine the electrohydrodynamic coefficients.

5.4 Nonreciprocity across scale

Nonreciprocity can fade or survive coarse-graining. Work on active mixtures shows conditions under which microscopic mediated asymmetry can yield an effective equilibrium description at larger scale, and conditions under which it persists and produces entropy, demixing, dynamic patterns, or state transitions. ([Dinelli et al. 2023](#); [Mohite and Rieger 2026](#); [Lee et al. 2026](#))

ASTRA should therefore record reciprocity at three levels:

- microscopic or agent-level coupling;
- mesoscopic coarse-grained constitutive law;
- whole-system conservation and entropy closure.

A statement such as “action-reaction is broken” is incomplete unless it names the level and the omitted environment.

6. Arrested coarsening and dynamic topology

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6.1 Static arrest versus dynamic arrest

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A static aggregate can stop changing because it reached an equilibrium, a glassy state, or a kinetic trap. A dynamically arrested cluster state is different: a characteristic scale remains bounded or statistically stationary while mergers, fragmentation, exchange, and reorganization continue.

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This operational formulation is **Proposed**. Let $\ell(t)$ be a declared characteristic-scale process and $\Gamma_{\text{turn}}(t)$ a microscopic turnover rate. An observation is consistent with dynamic arrest over a declared window only when:

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1. $\ell(t)$ is bounded within preregistered physical limits or statistically stationary under a preregistered assessment across the declared ensemble; and
2. turnover remains resolved above a preregistered positive threshold.

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The record must state the window, ensemble, sampling resolution, and noise treatment. A long-window trend statistic may diagnose unresolved drift, but vanishing logarithmic slope alone is neither necessary nor sufficient. For example, $\ell(t) = \log(1 + t)$ is unbounded although its logarithmic slope tends to zero, whereas $\ell(t) = 2 + \sin t$ is bounded although its logarithmic derivative does not converge. These conditions classify observations; they do not establish a universal law.

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6.2 Topology as an ensemble rather than one graph

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When clusters continually split and reform, one static graph is a poor description. Let $\mathcal{G}(t)$ be the instantaneous contact or interaction graph and let $P(\mathcal{G} \mid \mu, u)$ be the mode-conditioned graph ensemble. The scientific object may be the stationary or slowly evolving distribution over graph motifs rather

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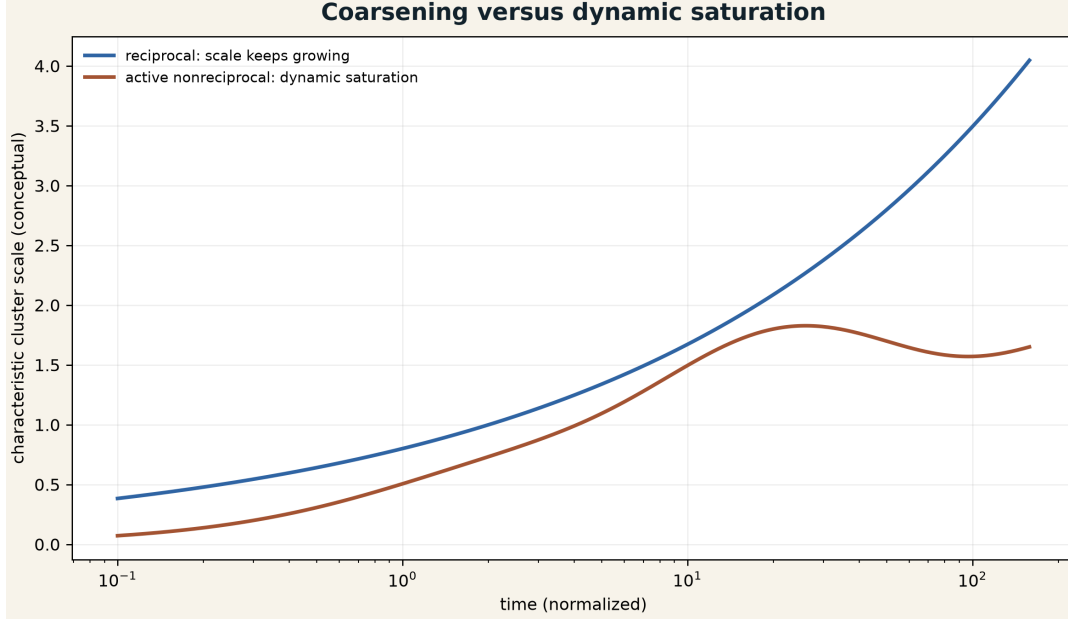


Figure 6: [MODEL] Conceptual distinction between reciprocal coarsening and a dynamically saturated nonreciprocal state. The trajectories are explanatory and are not fitted to Hara et al. Creator: ASTRA / Jacko T. Source: original model figure. License: CC BY 4.0.

than a single recovered topology.

Useful observables include:

$$P(s), \quad P(k), \quad \tau_{\text{edge}}, \quad \Gamma_{\text{merge}}, \quad \Gamma_{\text{split}}, \quad \Phi_{\text{motif}}, \quad \dot{S}_i,$$

where s is cluster size, k degree, τ_{edge} edge lifetime, and Φ_{motif} motif-transition flux. A candidate nonreciprocal model should predict those jointly, not merely reproduce one snapshot.

6.3 Planetary relevance is conditional

Planetary transport can contain directed effective couplings: sediment settling with active biological transport, chemistry coupled to flow, charge-separated dusty plasmas, rotating magnetized fluids, and reaction-diffusion

networks. The colloid paper does not validate any planetary application. 646
 It supplies a calibration standard for what a directed edge, environment 647
 closure, and dynamic graph ensemble look like in a controlled experiment. 648

7. Self-rewriting mechanochemical interfaces 649

7.1 The Cu3Pt result 650

Redondo and colleagues deposited Cu3Pt intermetallic thin films on a NiTi 651
 shape-memory substrate. The substrate’s martensite-austenite transforma- 652
 tion imposed approximately +0.80% in-plane tension or −0.99% compression. 653
 In 0.5 M sulfuric acid, the compressed film reached a reported 855 mV at 654
 1.0 mA cm⁻², compared with 856 mV for the pure-Pt thin-film comparison; 655
 tension reduced the value to 840 mV. Electrochemical cycling selectively de- 656
 pleted Cu, producing a reported 5-10 nm Pt-enriched surface layer. ([Redondo 657
 et al. 2026](#)) 658

The result is important, but the phrase “same performance” must remain 659
 bounded. It means near-equal potential at one declared current density in 660
 the reported thin-film acidic electrochemical test. It does not mean equal 661
 durability, Pt mass activity across all currents, membrane-electrode-assembly 662
 behavior, start-stop tolerance, poisoning resistance, or system-level cost. 663

7.2 Fast and slow edge state 664

A useful catalyst edge state is 665

$$b_e = (\varepsilon, c_{\text{surf}}, h_{\text{Pt}}, E_d, \Gamma_O, \Gamma_{OH}, \rho_{\text{defect}}, \zeta_{\text{rough}}, \lambda_{\text{hydration}}, \chi_{\text{poison}}, N_{\text{cycle}}).$$

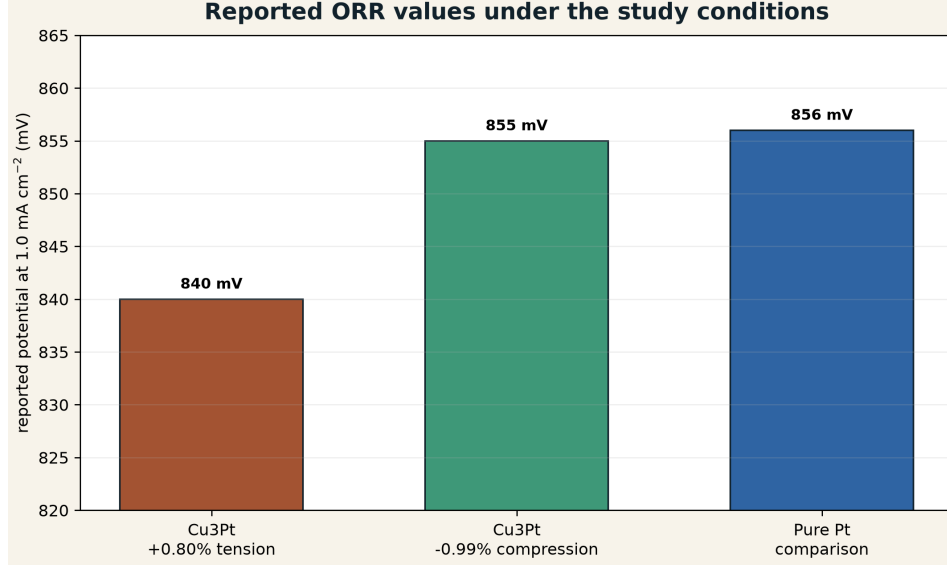


Figure 7: [OBSERVATION SUMMARY] Reported ORR potentials under the study conditions. The plot redraws three values from Redondo et al. and does not reproduce a journal figure. The abstract does not provide uncertainty bars for these values. It does not establish full-cell power density, durability, manufacturing cost, or commercial parity. Creator: ASTRA / Jacko T. Source data: Redondo et al. License for this redrawn chart: CC BY 4.0.

666 The ORR current is schematically

$$j_{\text{ORR}} = G_{\text{ORR}}(\eta, T, c_{\text{O}_2}, b_e),$$

667 with an edge evolution law

$$\dot{b}_e = F_e(b_e, \varepsilon(t), j, T, c_{\text{O}_2}, c_{\text{electrolyte}}).$$

668 Strain may be partly reversible on the experimental timescale. Surface com-
 669 position, Pt enrichment, roughness, and defects may evolve irreversibly or
 670 with much longer relaxation. The observed response can be decomposed con-
 671 ceptually as

$$\Delta j = \Delta j_{\text{elastic}} + \Delta j_{\text{chemical}} + \Delta j_{\text{cross}} + \Delta j_{\text{nuisance}}.$$

The cross term matters because strain and dealloyed surface structure need not act independently.

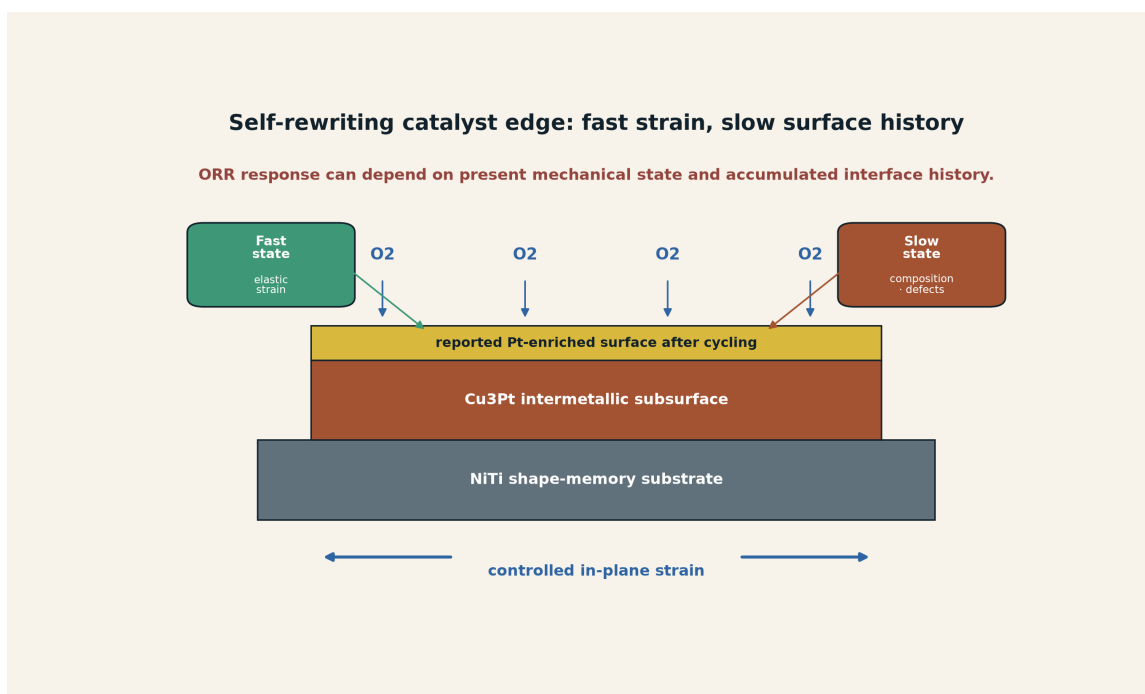


Figure 8: [MODEL] A self-rewriting catalytic edge. Fast controlled strain and slower cycling-induced composition change jointly determine the ORR interface. The layer thickness and geometry are schematic; only the reported 5-10 nm Pt-enriched layer is numerically sourced. Creator: ASTRA / Jacko T. Source: original model informed by Redondo et al. License: CC BY 4.0.

7.3 Novelty boundary

Elastic strain engineering of catalytic activity predates this study, including experiments using NiTi substrates and work showing that strain and adsorbate coverage can jointly modify adsorption energies and rate-limiting steps. (Monclús et al. 2025; Martínez-Alonso et al. 2025)

The strongest defensible novelty is therefore not “strain affects catalysis” in

general. It is the reported integration of controlled sub-percent strain, Cu₃Pt ORR activity, selective dealloying, a Pt-enriched working surface, and a pure-Pt thin-film comparison that reaches near-equal potential at the declared current density with one quarter of the Pt content.

7.4 Required promotion experiment

A high-information follow-up should vary strain and dealloying independently. One design is a factorial protocol with:

- compression, tension, and zero-strain controls;
- pre-dealloyed and undealloyed surfaces;
- fixed Pt mass and independently measured electrochemical surface area;
- operando strain, composition, and adsorbate measurements;
- long-duration cycling and thermal cycling;
- full current-voltage curves, Tafel behavior, impedance, mass activity, and dissolution rates;
- a membrane-electrode-assembly test held out from thin-film calibration.

The self-rewriting-edge model is weakened if strain loses predictive value after surface area and composition are controlled, or if the effect disappears under realistic operation.

Part III - Active support, visibility, and sector- complete instruments

8. Mode-resolved active support

8.1 Why nominal input is too coarse

A source can be present without coupling effectively. The active-support draft was motivated by three heterogeneous cases: a flying-focus accelerator in which the useful overlap moves through spacetime; radiofrequency disturbance of bird orientation in which waveform envelope matters; and a catalyst lead in which different geometric measures may control opposite operating modes. (Jacko T. 2026g; Arrowsmith et al. 2026; Kavokin et al. 2026)

The general correction is

source $\rightarrow$ waveform/control $\rightarrow$ active support
 $\rightarrow$ local state change $\rightarrow$ system output
 $\rightarrow$ residue $\rightarrow$ prediction.

Total energy, peak field, total catalyst mass, or nominal frequency may fail to identify the effective interaction.

8.2 Support kernel

For operating mode μ and control u , let

$$a_{\mu,u}(\mathbf{r}, t, \nu) \in [0, 1]$$

713 be a dimensionless support weight. Let $R_{\mu,u}$ be a local response density with
 714 declared units and measure. Then

$$Y_{\mu,u} = \int_{\Omega_r} \int_T \int_{\Omega_\nu} a_{\mu,u}(\mathbf{r}, t, \nu) R_{\mu,u}(X, b_E; \mathbf{r}, t, \nu) d\nu dt d\mathbf{r}.$$

715 This is a bookkeeping interface, not a universal constitutive law. The appli-
 716 cation must state the spatial dimension, coordinates, time window, spectral
 717 measure, units of R , normalization of a , threshold rule, and boundary treat-
 718 ment.

719 A thresholded support is

$$\mathcal{A}_{\mu,u}(\vartheta) = \{(\mathbf{r}, t, \nu) : a_{\mu,u} > \vartheta\}.$$

720 For any $c > 0$ such that ca remains in the declared admissible support-weight
 721 domain (in particular, $0 \leq ca \leq 1$ here), the transformation $a \mapsto ca$, $R \mapsto$
 722 R/c leaves Y unchanged. That restricted gauge freedom means an active-
 723 support map is not uniquely identified without a normalization convention or
 724 independent measurement.

725 **8.3 Moving-front coordinate**

726 For a front with speed v_f , interaction time τ_{int} , and interaction length ℓ_{int} ,

$$\Xi = \frac{v_f \tau_{\text{int}}}{\ell_{\text{int}}}.$$

727 $\Xi \ll 1$ means the front moves little over the local interaction span; $\Xi \sim 1$
 728 indicates comparable scales; $\Xi \gg 1$ means the front traverses the span rapidly.

A peak near $\Xi = 1$ is a proposed hypothesis, not a theorem. It must be tested by independently varying or measuring the three scales rather than tuning ℓ_{int} after observing the response.

8.4 Active support in the two new cases

In the colloid experiment, the active support is not the entire fluid volume. It is the particle-dependent electrohydrodynamic flow field, the particle pair geometry, the electrode gap, and the time-dependent field condition that sustains directed pair interactions.

In the catalyst experiment, active support includes the Pt-enriched reaction surface and the strained subsurface region that modifies adsorption energetics. The relevant measure may be electrochemically active area, interface perimeter, surface composition, or a weighted combination rather than total film volume.

8.5 Support falsifier

A support assignment becomes scientific when a selective perturbation changes the output as predicted while matched total input is preserved. Examples include:

- scramble the focal trajectory while holding laser energy fixed;
- change RF envelope while matching peak, RMS, and mean power in separate controls;
- vary catalyst geometry or strain while holding Pt mass and composition fixed;
- mask or restore a predicted spatial reaction zone;
- add a second detector port expected to see the omitted support.

A support hypothesis should be demoted if a different support map predicts held-out outcomes equally well with fewer assumptions, or if the proposed support can be arbitrarily redrawn without changing likelihood.

9. Sector-complete instruments

9.1 Correct observation equations

The Sector-Complete Instrument alpha repaired a blocking mathematical error. A trace of a commutator cannot serve as a generic measurement expectation because

$$\text{Tr}[A, B] = 0$$

under ordinary finite-dimensional conditions. The corrected quantum expectation is

$$\langle O_j \rangle = \text{Tr} [O_j \mathcal{E}_{\Gamma, u}(\rho)] ,$$

or, for detector outcome d ,

$$p(d \mid \rho, \Gamma, u) = \text{Tr} [M_d \mathcal{E}_{\Gamma, u}(\rho)] , \quad M_d \succeq 0, \quad \sum_d M_d = I.$$

$\mathcal{E}$ is completely positive and trace preserving when no postselection occurs. A selected outcome branch uses a trace-nonincreasing quantum instrument. Counts, intensities, voltages, forces, and reconstructed parameters require their own unit-bearing likelihoods; a universal additive-noise term is not ade-

quate. (Jacko T. 2026c)

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9.2 Observational quotient

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Let $\mathcal{K}$ be a declared candidate-generator set and π an observation protocol.

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Define

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$$K_i \sim_{\pi} K_j \iff p(D \mid K_i, \pi) = p(D \mid K_j, \pi).$$

The experiment certifies only

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$$\mathcal{K} / \sim_{\pi}.$$

This is the precise form of the ASTRA warning that more measurements in the same basis may reproduce the same ambiguity. A protocol becomes sector complete only relative to $\mathcal{K}$: every plausible output sector has a measurement or quantitative bound, detector cross-talk and loss are modeled, unresolved classes are reported, and an out-of-set goodness-of-fit test remains active.

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9.3 The frozen four-generator benchmark

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The alpha module uses four synthetic generators:

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$$\{\text{reflect}, \text{absorb}, \text{local transmit}, \text{string transmit}\}.$$

A local detector sees only left, right, or no local signal. Its exact classes are

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$$\{\text{reflect}\}, \quad \{\text{absorb}, \text{string transmit}\}, \quad \{\text{local transmit}\}.$$

Adding string, environment, and interface-state observations separates the four candidates inside the frozen model. At 2% symmetric detector confusion, the released benchmark reports Fisher rank 2 versus 3 on the three-dimensional mixture simplex, mutual information 1.343487 versus 1.853974 bits, and classification accuracy 0.7525 versus 1.0000 for local versus expanded protocols. These are synthetic results under a declared candidate set, not evidence for real duality defects or hidden matter. ([Jacko T. 2026c](#))

9.4 Application to nonreciprocal colloids

A particle-only observation basis can suggest an unexplained action-reaction residual. A sector-complete colloid instrument should include or bound:

- particle trajectories and pair geometry;
- local fluid velocity or flow proxies;
- electric-field amplitude and phase;
- electrode current and boundary conditions;
- particle-size distribution and surface state;
- temperature and viscosity;
- environmental momentum and energy flux.

If the environment channels are unmeasured, the result may still establish effective nonreciprocal particle dynamics, but it cannot certify a closed-system momentum anomaly.

9.5 Application to the catalyst

A current-voltage curve alone cannot separate strain, dealloying, surface area, coverage, roughness, and dissolution. A sector-complete catalyst instrument should measure or bound:

- in-plane strain during operation; 805
- bulk and surface composition; 806
- Pt-rich layer thickness; 807
- active surface area; 808
- O, OH, and water coverage proxies where feasible; 809
- Cu dissolution products; 810
- morphology and defect evolution; 811
- current, potential, impedance, temperature, and mass transport. 812

The unresolved quotient should be reported explicitly. If strain and Pt enrichment remain observationally equivalent under the available protocol, the model should not claim that one alone caused the improvement. 813
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10. Operator-aware visibility and sampling 816

10.1 The visibility composition 817

The Cosmic Visibility and Sampling Framework formalizes the route from a source field $\mathcal{R}_H$ to observed data: 818
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$$\mathcal{V}_m = \mathcal{O}_m \circ \mathcal{S}_m \circ \mathcal{P}_m \circ \mathcal{T}_m,$$

$$\mathbf{y}_m = \mathcal{V}_m[\mathcal{R}_H] + \mathbf{f}_m + \varepsilon_m.$$

$\mathcal{T}$ is transduction, $\mathcal{P}$ propagation, $\mathcal{S}$ archive or sampling, $\mathcal{O}$ detector response, $\mathbf{f}$ foregrounds, and ε noise. (Jacko T. 2026e) 820
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When a scalar factorization is justified at a declared resolution, 822

$$\eta_m(q; \psi) = \eta_{\text{prod}} \eta_{\text{trans}} \eta_{\text{prop}} \eta_{\text{arch}} \eta_{\text{samp}} \eta_{\text{obs}}.$$

The product is conditional bookkeeping, not a universal independence claim. Coupled or history-dependent operators may require integral kernels or state-space models.

10.2 Source-visibility degeneracy

If data depend primarily on a product such as

$$\Phi \propto \frac{b}{\tau} B^2 \ell_{\text{coh}} f_{\text{vol}},$$

then a null constrains a manifold in source lifetime, branching fraction, magnetic field, coherence length, and volume fraction. It does not measure one parameter in isolation. This is the visibility analogue of topology non-identifiability.

A model should record the Fisher or sensitivity matrix

$$F_{ab} = \mathbb{E} [\partial_a \log p(D \mid \vartheta) \partial_b \log p(D \mid \vartheta)].$$

Null directions indicate parameter combinations that remain unidentified. Mutual information can guide a frozen synthetic design, but it is prior dependent and not itself a scientific certificate.

10.3 Archive and sampling bias

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The Mars meteorite case demonstrates the opposite problem. The observed
collection is conditioned by melt production, impact excavation, ejection, in-
terplanetary transfer, atmospheric survival, terrestrial preservation, human
recovery, and classification. A gap in the collection does not directly imply a
gap in Martian history. (Seal et al. 2026; Jacko T. 2026e)

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The same logic applies to fossils, archaeological sites, exoplanet catalogs, tran-
sient surveys, and laboratory yield. The expected observable count or sig-
nal must include the sampling operator before an absence is assigned veto
strength.

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10.4 Unified operator stack

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11. The SPPT Bridge Protocol

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11.1 Five fail-closed gates

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The repository’s Bridge Protocol prototype offers the most concrete integra-
tion path:

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Conservation Contract. Declare incidence, stoichiometry, sources, sinks,
units, and weighted invariants. Verify static and dynamic residuals.

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Thermodynamic Ledger. Record energy and entropy terms once, reject du-
plicate IDs or nonfinite values, and apply nonnegative production checks on
their stated domains.

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Observational Equivalence. Compute finite transfer signatures, pole/zero
structure, controllability/observability ranks, or another domain-appropriate

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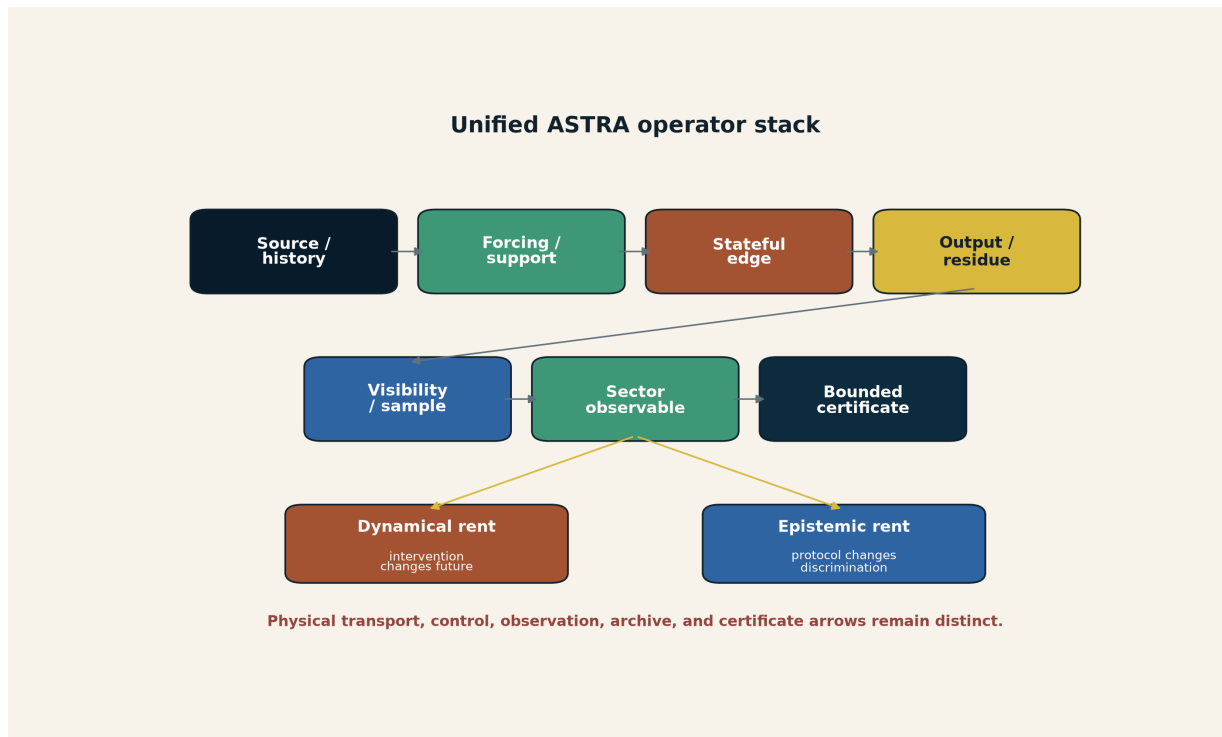


Figure 9: [MODEL] Unified ASTRA operator stack. The lower modules identify which successor method audits each segment. The stack is a typed inference architecture, not a claim that all domains share one carrier or constitutive law. Creator: ASTRA / Jacko T. Source: original vector model. License: CC BY 4.0.

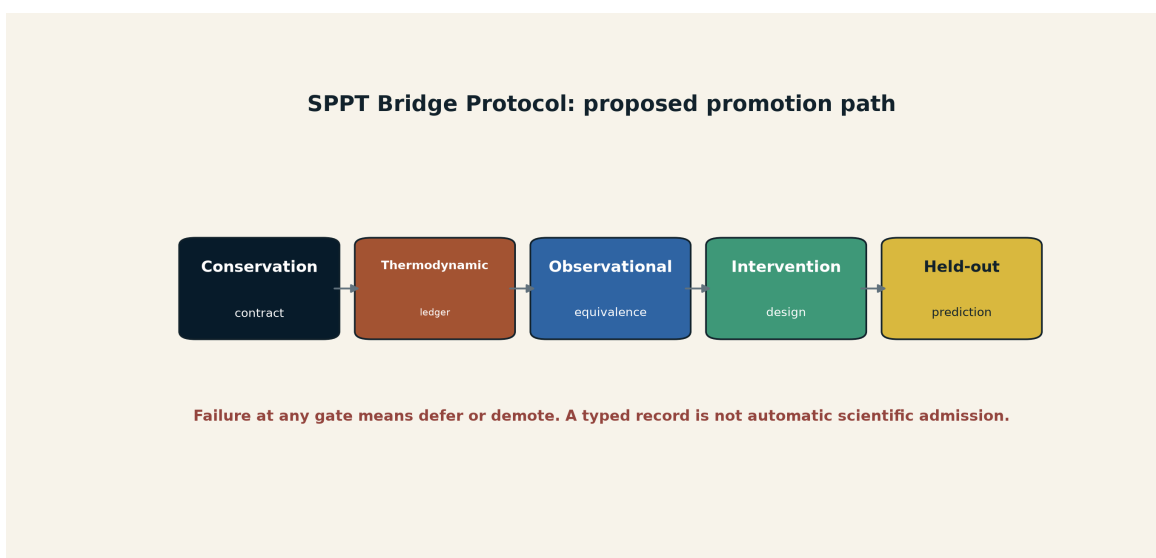


Figure 10: [MODEL] Candidate promotion path. Failure at any gate produces defer or demote rather than reinterpretation. Creator: ASTRA / Jacko T. Source: original vector model based on the repository prototype. License: CC BY 4.0.

equivalence diagnostic. A finite signature is not proof of rational transfer
equality unless a theorem or canonical representation closes the gap.

Intervention Design. Select forcing, ports, frequencies, support perturbations, or sector measurements that maximize response separation subject to explicit cost and safety constraints.

Calibrated Prediction Audit. Fit model means before the calibration split, estimate scales only on calibration data, and score held-out data with log score, CRPS, interval coverage, posterior predictive checks, or simulation-based calibration as appropriate.

11.2 Utility without hidden units

An intervention score such as

$$U(\pi) = \frac{I(K; D_\pi) - \lambda R(\pi)}{C(\pi) + \mu S(\pi)}$$

is meaningless unless information, redundancy, cost, and safety are normalized and the utility weights are declared. The candidate requires sensitivity analysis over λ , μ , priors, cost units, and stopping rules.

11.3 Out-of-set rejection

A model-selection system must be able to reject its candidate family. Good classification among four wrong models is not scientific success. The Sector-Complete alpha's out-of-set hybrid control and the Bridge Protocol's posterior predictive diagnostics should be combined into one mandatory gate:

select within family only after family adequacy is not rejected.

12. Dual rent and local-to-global certificates

12.1 Dynamical rent

For a seam or edge intervention Γ relative to reference Γ_0 ,

$$R_{\text{dyn}}(\Gamma) = d[P(Y \mid do(\Gamma)), P(Y \mid do(\Gamma_0))].$$

The distance d , intervention, held variables, and uncertainty model must be declared. Catalyst strain and nonreciprocal pair coupling are examples of potential high dynamical rent because changing the interface changes the future response.

12.2 Epistemic rent

For candidate generator K and protocol π ,

$$R_{\text{epi}}(\Gamma, \pi) = I(K; D \mid \Gamma, \pi) - I(K; D \mid \Gamma_0, \pi).$$

A new detector sector, forcing frequency, cosmic transducer, or calibrated proxy can increase causal discrimination even if it barely changes the physical system.

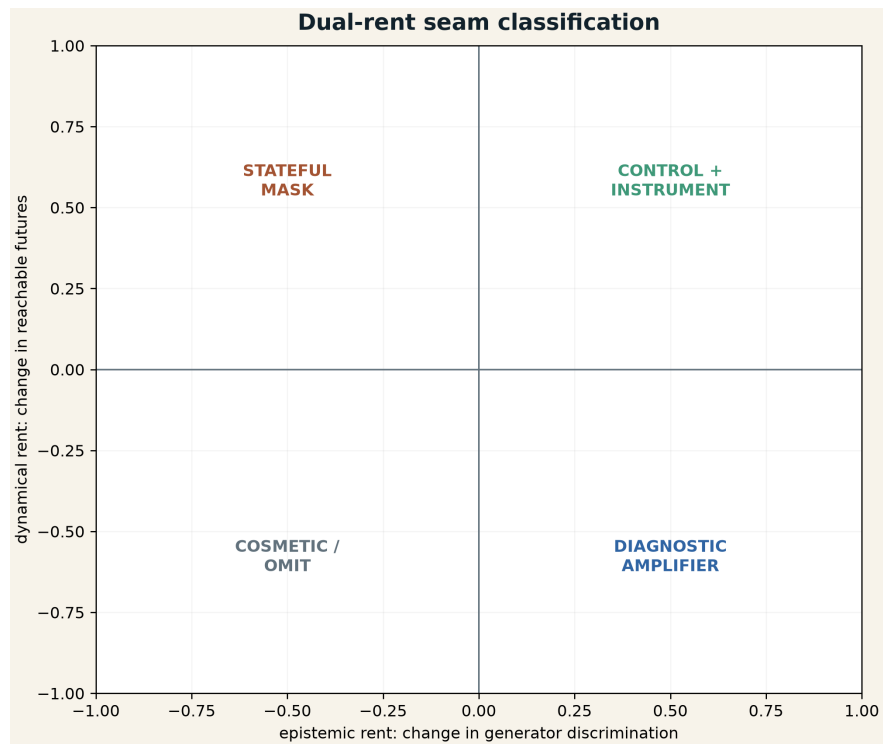


Figure 11: [MODEL] Dual-rent classification. A seam may change the future, improve identifiability, do both, or do neither. Coordinates are conceptual, not measured values for the cited experiments. Creator: ASTRA / Jacko T. Source: original vector model. License: CC BY 4.0.

12.3 Local-to-global stack

The candidate retains the five-level certificate stack from the Earth supplement:

$$\mathcal{C}_{\text{L2G}} = (C_{\text{local}}, C_{\text{fiber}}, C_{\infty}, C_{\text{arith}}, C_{\text{formal}}).$$

Certificate	Inspects	Does not automatically establish
local	differential, constitutive response, local stability	global uniqueness, closure, distant collisions
fiber	complete preimage or candidate structure	behavior at omitted boundaries
infinity/closure	escape routes, properness, omitted reservoirs	arithmetic or implementation correctness
arithmetic	prime reductions, valuations, extension behavior	characteristic-zero result without lifting
formal	exact identities, proof objects, tests, hashes	empirical adequacy or historical truth

The stack is directly relevant to the two new studies. A reported ORR potential is a local performance certificate, not a global fuel-cell certificate. Particle-pair drift is a local reduced-interaction certificate, not a closed-system law. The correct response is not to weaken the local result, but to stop the claim at the level it actually certifies.

Part IV - Calibration cases and cross-domain synthesis 897 thesis 898

13. Calibration case: nonreciprocal active colloids 899 loids 900

13.1 What the primary record supports 901

Hara and colleagues report a bidisperse suspension of polystyrene colloids under an alternating electric field. The particles had radii of approximately 1 and 1.5 micrometers and were confined in water between transparent ITO-coated electrodes. Size-dependent electrohydrodynamic flows produced asymmetric effective attraction. Differently sized pairs acquired a front-back polarity and moved as self-propelled units. At larger scale, more than 10,000 particles were observed for over an hour in clusters that repeatedly fragmented, reorganized, and reformed rather than coarsening into static aggregates. Agent-based simulations reproduced the qualitative dynamics and identified nonreciprocal pair propulsion as the minimal model ingredient for persistent clustering. ([Hara et al. 2026](#)) 902
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Scientific status: Established. Scope: the reported driven colloidal platform. 913
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Best ordinary interpretation: A nonequilibrium, field-driven, fluid-mediated reduced subsystem with effective nonreciprocal interactions. 915
916

Rejected interpretation: Fundamental creation of momentum or universal failure of Newton's third law in the closed system. 917
918

13.2 Scientific advancement

The experiment advances the field in three ways.

First, it scales a controllable nonreciprocal interaction beyond small clusters to a dense assembly of more than ten thousand particles.

Second, it demonstrates **arrested coarsening through activity**: asymmetry does not merely translate isolated pairs; it maintains a cluster-scale turnover process.

Third, it links a microscopic asymmetry to a mesoscopic material state that is neither an equilibrium crystal nor simple unbounded aggregation.

The result belongs to a broader convergence in nonreciprocal active matter, including demixing in flocking mixtures, thermodynamic accounting for nonreciprocal particle-field systems, and topological descriptions of state transitions in living matter. (Mohite and Rieger 2026; Lee et al. 2026)

13.3 ASTRA bridge

The safe ASTRA contribution is **directed edge state with environment closure**. The experiment shows that a graph with symmetric adjacency but asymmetric couplings can possess a different attractor structure from its reciprocal counterpart. It also shows why a particle-only graph is incomplete: the interaction is mediated by field-driven fluid flow.

A candidate record should include:

$$\mathcal{R}_{\text{colloid}} = (B_0, \omega, d_{\text{gap}}, \eta_f, T, R_L, R_S, K_{LS}, K_{SL}, \rho, \phi, \Gamma_{\text{split}}, \Gamma_{\text{merge}}, \dot{S}_i, \mathcal{L}_{\text{env}}).$$

No quantity should be imported into planetary or biological models without a domain-specific law.

13.4 Highest-information next tests

The most useful next experiments are not simply larger assemblies. They should independently vary the asymmetry and mediator:

- particle size ratio at fixed area fraction;
- field amplitude and frequency;
- electrode separation and boundary condition;
- fluid viscosity and conductivity;
- monodisperse reciprocal controls;
- direct or proxy flow-field measurement;
- externally imposed reciprocal attraction matched in static strength;
- sudden field removal and reversal;
- held-out prediction of cluster-size distribution and turnover rates.

The model should be demoted if a reciprocal interaction plus unmodeled heterogeneity predicts the same held-out dynamics, or if the environment ledger cannot account for the observed directed motion.

14. Calibration case: strain-engineered Cu₃Pt ORR catalysis

14.1 What the primary record supports

The primary paper reports controlled elastic strain in Cu₃Pt intermetallic thin films on a NiTi substrate, ORR measurements in acidic electrolyte, a

compressed-film potential of 855 mV at 1.0 mA cm⁻², a pure-Pt comparison of 856 mV, a tensile value of 840 mV, and cycling-induced selective Cu dealloying that leaves a 5-10 nm Pt-enriched surface. ([Redondo et al. 2026](#))

Scientific status: Established. Scope: the reported thin-film electrochemical study.

Best ordinary interpretation: Strain and evolving surface composition jointly alter adsorption energetics and ORR response.

Open: long-term strain stability, full-cell durability, industrial fabrication, total cost, and system-level performance.

14.2 Scientific advancement

The result converts strain from an incidental materials variable into a controlled operating coordinate. It also demonstrates that the working catalyst is not identical to the as-deposited catalyst: electrochemical operation changes the surface. That makes history part of the functional state.

The strongest general lesson is:

same nominal alloy+different strain/history → different constitutive response.

14.3 ASTRA bridge

The study supplies a concrete **self-rewriting edge**. The relevant state includes reversible elastic deformation and slower composition or defect evolution. A correct model should treat cycling as an intervention that changes the edge, not as repeated observation of a fixed object.

14.4 Highest-information next tests981

The promotion program should include full-cell tests, but the immediate mech-982
anistic priority is a strain-composition orthogonalization experiment. If strain983
and dealloying cannot be varied independently, the paper’s causal interpreta-984
tion remains partly entangled.985

A useful rejection criterion is:986

After controlling active surface area, Pt enrichment, roughness,987
mass transport, and temperature, the strain coordinate fails to988
predict held-out ORR response or relaxes too rapidly to remain989
operationally relevant.990

15. Calibration portfolio from the current991
project992

The present update does not reproduce every prior report in full. It records993
what each case contributes to the integrated method.994

ASTRA			
Case	Domain result	contribution	Main limit
Flying-focus wakefield	moving focus	active support	not a
	extends phase	and velocity	demonstrated
	matching beyond	matching	100 GeV stage
	conventional		
	dephasing in a 7		
	mm experiment		

Case	Domain result	ASTRA	
		contribution	Main limit
Pulsed RF bird study	selected modulated fields disrupted group orientation more than matched continuous conditions	waveform-resolved support; receptor uncertainty	does not identify receptor or human-health effect
Sunlight-pumped SPDC	sunlight can pump a filtered nonlinear apparatus that creates local entanglement	degree-of-freedom selection and transduction	sunlight did not arrive carrying the measured pairs
Duality defects	local excitations can convert to string/defect sectors in specified models	sector blindness and observable completeness	not a cosmological mirror world
Clavina photonics	route, timing, feedforward, and reusable nonlinear modules instantiate different tasks	architecture/control state	not fault-tolerant universal computation

Case	Domain result	ASTRA	
		contribution	Main limit
Fermium spectroscopy	calibrated electronic spectra infer hidden nuclear moments	model-mediated proxy measurement	not direct nuclear imaging
XENONnT null	constrains specified mass-coupling- response regions	bounded veto manifold and neutrino-fog equivalence	does not exclude dark matter as a whole
Cosmic filament conversion	hidden decay products may become electromagnetic through cosmic fields	natural transducer and visibility kernel	conditional model, no graviton detection
Mars meteorite	one sample fills a collection-age gap	selective archive and sampler	source reservoir remains nonunique
Galaxy-spin memory	present spins correlate with reconstructed primordial tidal structure	primordial residue channel	does not identify a dark-matter particle

Case	Domain result	ASTRA	
		contribution	Main limit
Higher-dimensional Jacobian counterexample	local nonsingularity does not force global injectivity for $n \geq 3$; plane case remains open	local-to-global certificate discipline	mathematical calibration, not cosmology
Accelerating-wave equation	a 2023 theory models prescribed time-varying wave speed and claims a positive-time branch	temporal-interface, frame, branch, and global-ledger audit	theoretical; not a universal arrow-of-time experiment

995 The cases are methodologically related, not physically unified. Primary
 996 records for the quantum-transduction cases include the sunlight SPDC
 997 preprint, the duality-defect preprint, the Clavina architecture preprint, and
 998 the fermium spectroscopy preprint. ([Li et al. 2026](#); [Ueda et al. 2026](#); [Yu et al.](#)
 999 [2026](#); [Urquiza-González et al. 2026](#))

15.1 Temporal-interface calibration: accelerating waves

The Brighter Side article published on 9 August 2026 is a discovery lead, not the primary scientific record. The underlying paper was published in *Optica* in October 2023. (Shavit 2026; Koivurova, Robson, and Ornigotti 2023)

The paper derives its accelerating-wave equation by composing first-order characteristic operators. In one spatial dimension, the resulting expression is

$$\partial_t^2 f = c(t)^2 \partial_x^2 f - \dot{c}(t) \partial_x f$$

for a wave speed prescribed as a function of time. The extra term records the change in the propagation law. For electromagnetic applications the authors write $c(t) = c_0/n(t)$, introduce an intrinsic time $t' = \int n(t)^{-1} dt$, and interpret several observer-dependent changes in frequency, wavelength, energy, and momentum through that reparametrization. (Koivurova, Robson, and Ornigotti 2023)

Three parts are already explained by the present ASTRA architecture.

1. **Stateful and controlled medium.** A time-varying refractive index is not passive background. The control schedule and medium state belong in the physical record.
2. **Reference fog.** Claims about unchanged or changed momentum depend on the declared observer, coordinate, and division between field and material degrees of freedom.
3. **Global closure.** Apparent local energy gain in a modulated medium must be closed by the pump and material that create the modulation. The paper itself makes this global-conservation qualification.

1022 What was not yet explicit in the v1.0.8 candidate is the **temporal-interface**
1023 **branch audit**. The new record asks:

- 1024 • Is the propagation law spatially varying, temporally varying, or evaluated
1025 along a trajectory?
- 1026 • Which external controller produces $n(t)$ or $c(t)$?
- 1027 • Which coordinate defines frequency, wavelength, phase, and momentum?
- 1028 • Which initial or boundary conditions select the reported solution branch?
- 1029 • What exactly is reversed in the proposed time-reversal operation: the field,
1030 medium, control schedule, environment, or only the sign of a coordinate?
- 1031 • Does the claim survive when the complete driven system is reversed rather
1032 than holding the pump history fixed?
- 1033 • Which measurement distinguishes a new physical law from an equivalent
1034 reformulation of Maxwell or standard wave dynamics?

1035 The field of time-varying photonics predates this paper and includes exper-
1036 imentally demonstrated temporal interfaces. A switched transmission-line
1037 metamaterial has produced temporal reflection and broadband frequency
1038 translation, with the switching apparatus supplying the time dependence.
1039 Those experiments show that temporal modulation is physical and measurable.
1040 They do not validate every relativistic or microscopic-arrow interpretation of
1041 the accelerating-wave equation. ([Galiffi et al. 2022](#); [Moussa et al. 2023](#))

1042 The evidence status is therefore:

- 1043 • **Established:** time-varying media and temporal-interface phenomena are
1044 real research domains; temporal reflection and frequency translation have
1045 been experimentally demonstrated.
- 1046 • **Established:** the 2023 paper derives and analyzes the displayed
1047 accelerating-wave equation as published theory.

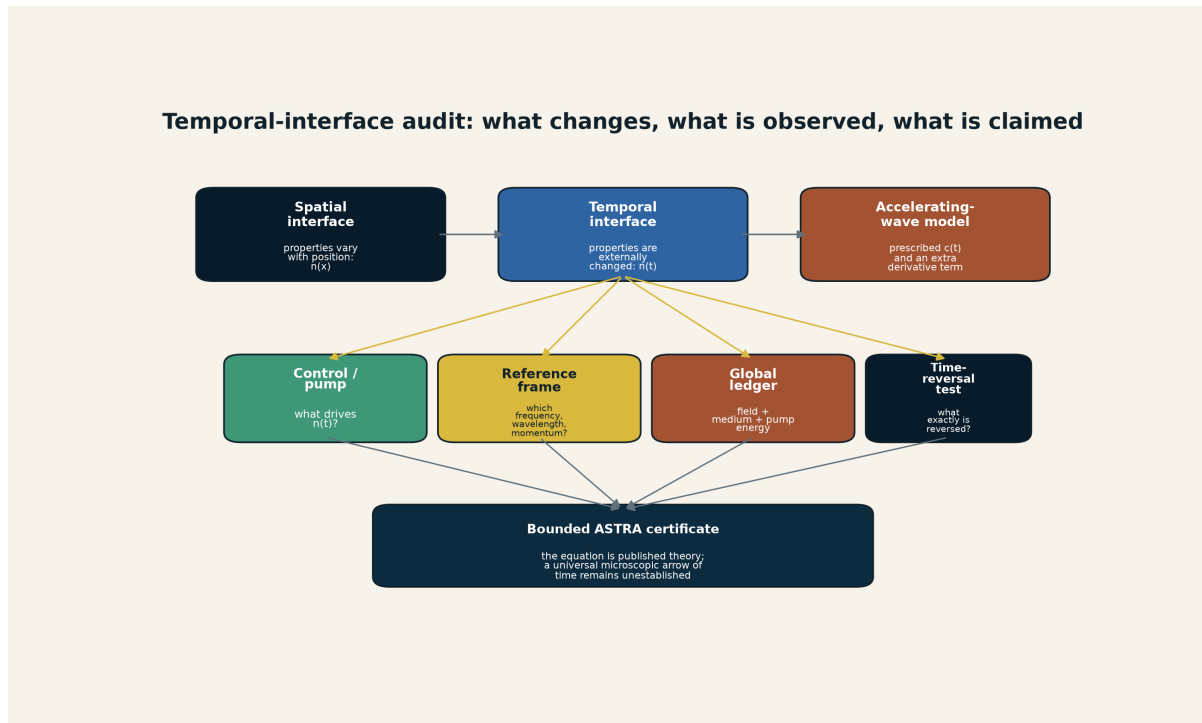


Figure 12: [MODEL] Temporal-interface audit for the accelerating-wave proposal. The diagram separates spatial interfaces, externally driven temporal interfaces, the prescribed $c(t)$ model, reference-frame choices, the pump and global ledger, and the stronger arrow-of-time interpretation. It does not depict an experiment or establish a universal arrow of time. Creator: ASTRA / Jacko T. Source: original synthesis based on Koivurova et al. (2023), Galiffi et al. (2022), and Moussa et al. (2023). License: CC BY 4.0.

- 1048 • **Plausible / model-dependent:** the intrinsic-time representation may be a
1049 useful continuous reformulation for selected wave problems.
- 1050 • **Open:** whether the formalism resolves the Abraham-Minkowski contro-
1051 versy generally.
- 1052 • **Open and unestablished:** whether the positive-time solution branch is a
1053 fundamental microscopic explanation of time's arrow.

1054 The most discriminating next step is not another conceptual illustration. It
1055 is a preregistered comparison in a driven temporal medium between the
1056 accelerating-wave prediction and standard Maxwell/Floquet models, using
1057 the same modulation schedule, losses, dispersion, pump energy, and detector
1058 model. A genuine advance must predict a measurable residual that the
1059 established formulations do not reproduce.

1060 **16. AEOF discipline for cross-domain novelty**

1061 Every proposed bridge should receive an AEOF record:

$$\text{AEOF} = (A, K, E, T, U, O, N, F, P, S),$$

1062 where A is the analogy, K the established kernel, E the standard equation, T
1063 the proposed new term, U units and domain, O observable, N null model, F
1064 falsifier, P prior art, and S evidence status. ([Jacko T. 2026d](#))

1065 For the two new studies:

Field	Nonreciprocal colloids	Cu ₃ Pt catalyst
analogy	directed effective edges keep structure active	operation rewrites the interface that performs it
established kernel	pair propulsion, persistent clusters	strain-dependent ORR values, Pt-enriched surface
standard equation	overdamped electrohydrodynamic particle/field dynamics	ORR kinetics, adsorption energetics, transport
proposed term	closure-conditioned nonreciprocity record	fast/slow edge-state decomposition
observable	trajectories, flow, cluster turnover, environment flux	strain, composition, current, potential, dissolution
null model	reciprocal interaction plus heterogeneity	composition/surface- area change without strain effect
falsifier	matched reciprocal model predicts held-out dynamics	strain loses predictive value after controls
prior art	active mixtures and nonreciprocal thermodynamics	strain engineering, coverage, d-band models
status	structural inference	structural inference

1066 This table is the correct novelty boundary. The new contribution is an ASTRA
 1067 integration and audit pattern, not the discovery of nonreciprocity or strain
 1068 catalysis.

1069 **Temporal-interface AEOF record**

Field	Accelerating-wave calibration
analogy	a changing medium acts like an effective spacetime for the wave
established kernel	published 2023 equation; broader temporal-interface experiments
standard equation	Maxwell or domain-specific wave equations with driven constitutive parameters
proposed term	$-\dot{c}(t) \partial_x f$ in the declared one-dimensional model
observable	phase, frequency translation, waveform, energy exchange, reference-frame invariants
null model	standard Maxwell/Floquet or transfer-matrix model with the same pump schedule
falsifier	no held-out residual unique to the accelerating-wave formulation
prior art	time-varying media, temporal scattering, photonic time crystals

Field	Accelerating-wave calibration
status	published theory; universal arrow-of-time interpretation open

Part V - Endogenous visibility and transformed cosmic archives 1070

17. The Endogenous Visibility Principle 1072

17.1 From fixed operators to source-coupled transducers 1073

The simplest visibility framework writes data as a fixed operator acting on a source. That is adequate when the intervening medium is stable or externally calibrated. It fails when the source deposits enough momentum, energy, ionization, radiation pressure, heat, or chemical change to alter the converter itself. 1074 1075 1076 1077 1078

Let $S(t)$ denote a hidden source and $b(t)$ the state of the transducing medium. The observation model is 1079 1080

$$D(t) = \mathcal{O}_\pi[\mathcal{V}_{b(t)}(S(t))] + \epsilon(t),$$

with 1081

$$\dot{b} = F(b, S, \mathcal{V}_b[S], u, \xi),$$

1082 where u is external control and ξ represents environmental forcing and nui-
1083 sance processes. The source and visibility state are jointly inferable only when
1084 the protocol contains enough independent information to distinguish them.

1085 Joint recovery of a source and an uncertain response operator is established
1086 inverse-problem territory, not a new ASTRA idea. Blind deconvolution has
1087 long treated a spectrum together with an incompletely known spectrometer
1088 blur, and multichannel imaging work has explicitly solved instrumental de-
1089 convolution jointly with blind source separation, including in astrophysical
1090 settings. (O’Leary 2009; Jiang, Bobin, and Starck 2017) The candidate’s nar-
1091 rower contribution is the typed admission rule: when the transducer evolves
1092 dynamically, its state, closure ledger, observational quotient, and held-out in-
1093 terventions must be represented together. ASTRA does not claim priority for
1094 generic joint source/operator inference.

1095 The corresponding principle is:

1096 **When the source materially changes the medium that makes**
1097 **it observable, infer the source and medium together. A fixed**
1098 **visibility operator is not a valid certificate until backreaction**
1099 **has been bounded.**

1100 This is not a new field equation. It is an admission rule for forward models.

1101 **17.2 Four observational failure modes**

1102 **Source-shell conflation** occurs when the observed spectrum is attributed
1103 directly to the engine even though a dense envelope or scattering medium
1104 produces the defining features.

1105 **Trigger-basis blindness** occurs when a physical event lies outside the pass-

band, cadence, threshold, or alert logic of the survey that historically defined the class.

Self-detuning occurs when conversion changes the medium's resonance, suppressing the process before a linear calculation reaches its expected yield.

Catastrophic remapping occurs when destruction, transport, sorting, and reaccrusion expose material while breaking its original spatial and chronological context.

The four failures share an inverse-problem structure. They do not share one physical mechanism.

17.3 Backreaction-detuning diagnostic

For a resonant transducer, define the proposed dimensionless diagnostic

$$\Xi_{\text{br}} = \frac{|\delta\omega_{\text{medium}}|}{\Gamma_{\text{res}}},$$

where $\delta\omega_{\text{medium}}$ is the source-induced resonance shift and Γ_{res} the effective resonance width under the declared protocol. When $\Xi_{\text{br}} \ll 1$, a fixed linear response may be adequate. When $\Xi_{\text{br}} \gtrsim 1$, the medium can materially alter or terminate the process and must be evolved dynamically.

The number is a proposed audit coordinate, not a universal threshold theorem. The numerator and denominator are domain-specific and must be independently estimated.

18. MoM-BH*-1 and source-shell separation

18.1 Observation

JWST measured MoM-BH*-1 at spectroscopic redshift $z = 7.7569$, corresponding to about 660 million years after the Big Bang. The source has an extreme Balmer break with reported strength $7.7^{+2.3}_{-1.4}$, broad multi-peaked $H\beta$ emission, deep $H\beta$ and $H\gamma$ absorption, weak forbidden oxygen emission, and a compact rest-optical morphology. The break exceeds the published stellar-population limits used in the paper. (Naidu et al. 2026)

The authors model the source as an accreting black hole embedded in extremely dense, turbulent, nearly dust-free, Compton-thick hydrogen gas. Their fiducial Cloudy model uses $n_H = 10^{11} \text{ cm}^{-3}$, $N_H = 10^{25.8} \text{ cm}^{-2}$ and a turbulent velocity of 500 km s^{-1} . The physical structure is modeled as roughly 10-100 au in scale. The paper explicitly calls the construction highly simplified and intended to provide physical intuition. (Naidu et al. 2026)

18.2 Inference correction

A conventional broad-line mass estimate assumes that line width traces orbital velocity:

$$M_{\text{BH}} \sim \frac{R_{\text{BLR}} v_{\text{line}}^2}{G}.$$

If repeated scattering and dense-gas radiative transfer produce much of the width or multi-peaked structure, then v_{line} is not a pure kinematic observable. The envelope is part of the measurement operator. The paper concludes that masses of related little-red-dot sources may be overestimated by orders of

magnitude under naive local scaling relations. (Naidu et al. 2026)

1146

This does not eliminate the early-black-hole growth problem. It changes which part of the problem belongs to seed formation and which belongs to radiative-transfer inference.

1149

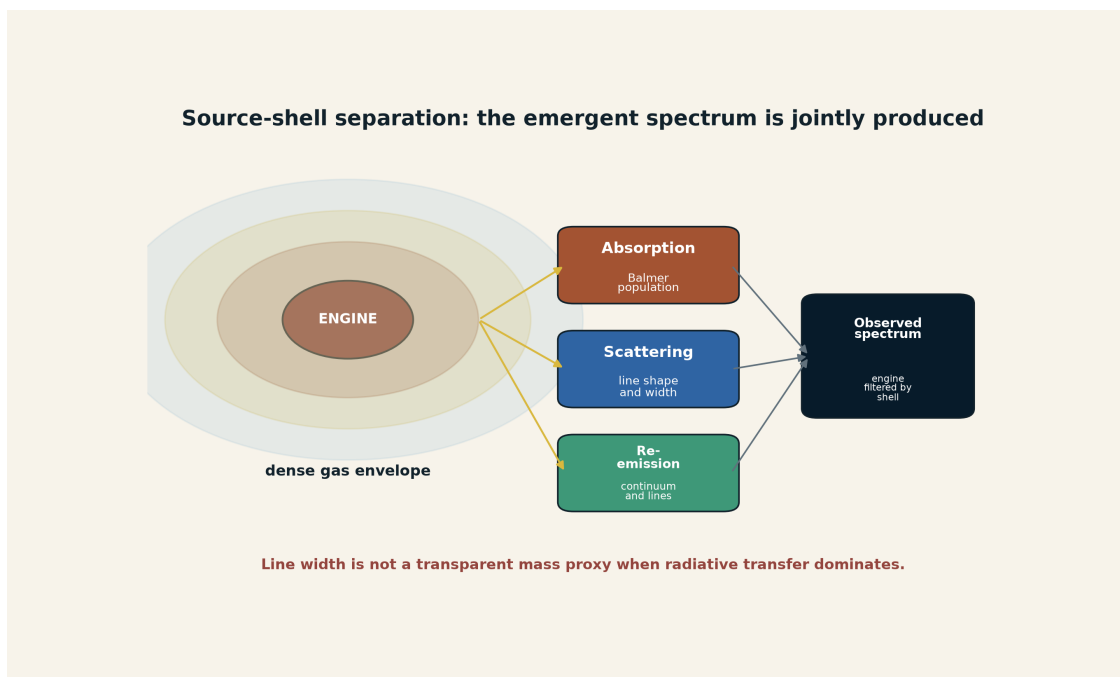


Figure 13: [MODEL] Source-shell separation for MoM-BH*-1. The detector sees radiation after absorption, emission, and scattering in a dense envelope. The diagram does not reproduce the JWST data or assert a unique envelope geometry. Creator: ASTRA / Jacko T. Source: original synthesis based on Naidu et al. License: CC BY 4.0.

18.3 Population status and falsifiers

1150

A template-based preprint reports 241 compact candidates whose rest-optical fits are more than 80% dominated by a black-hole-star template across approximately $z = 1.7-9.3$. That result is a candidate census, not spectroscopic confirmation of 241 gas-enshrouded black holes. (Weibel et al. 2026)

1154

The model would be strengthened by same-instrument variability, reverberation delays, X-ray constraints on Compton thickness, high-resolution Balmer

1156

1157 profiles, and host-galaxy kinematics that jointly fit one parameter set. It would
1158 be weakened if stellar or ordinary AGN models reproduce the complete contin-
1159 uum, absorption, line symmetry, variability, and infrared behavior with fewer
1160 assumptions.

1161 **Status:** observed spectrum **Established**; central accreting black hole
1162 **Strong inference**; dense gas envelope **Strong inference**; exact mass and
1163 accretion history **Open**.

1164 **19. EP250827b and cross-channel rescue**

1165 **19.1 Observation and classification**

1166 Einstein Probe detected EP250827b as a soft X-ray flash associated with the
1167 broad-line Type Ic supernova SN 2025wkm at $z = 0.1194$. The prompt X-
1168 ray luminosity was approximately $10^{45} \text{ erg s}^{-1}$, the event lasted more than
1169 1,000 s, and the peak energy was below 1.5 keV at 90% confidence. The opti-
1170 cal light curve was double-peaked and then remained near a 20-day bolomet-
1171 ric plateau, implying continuing energy injection beyond ordinary radioactive
1172 heating. ([Srinivasaragavan et al. 2026](#))

1173 The paper favors a long-lived magnetar, possibly accompanied by an accretion
1174 disk, whose winds mix, break out at approximately $0.35c$, and interact with
1175 circumstellar material extending to roughly 10^{13} cm . Those are model para-
1176 meters, not directly observed engine properties. Radio non-detection rules
1177 out a particular energetic on-axis jet only under the declared density and mi-
1178 crophysical assumptions. ([Srinivasaragavan et al. 2026](#))

19.2 The instrument network is the discovery object

1179

The original X-ray record was borderline. Rapid optical identification supplied 1180
a positional and chronological anchor; spectroscopy supplied the Type Ic-BL 1181
class; later photometry supplied the plateau; radio non-detection constrained 1182
selected jets. The event therefore became identifiable through a network: 1183

$$D_{\text{network}} = (D_X, D_{\text{optical}}, D_{\text{spectra}}, D_{\text{radio}}).$$

The important quantity is not simply the sum of exposures. Each channel adds 1184
a distinct causal coordinate. This is **cross-channel rescue**. 1185

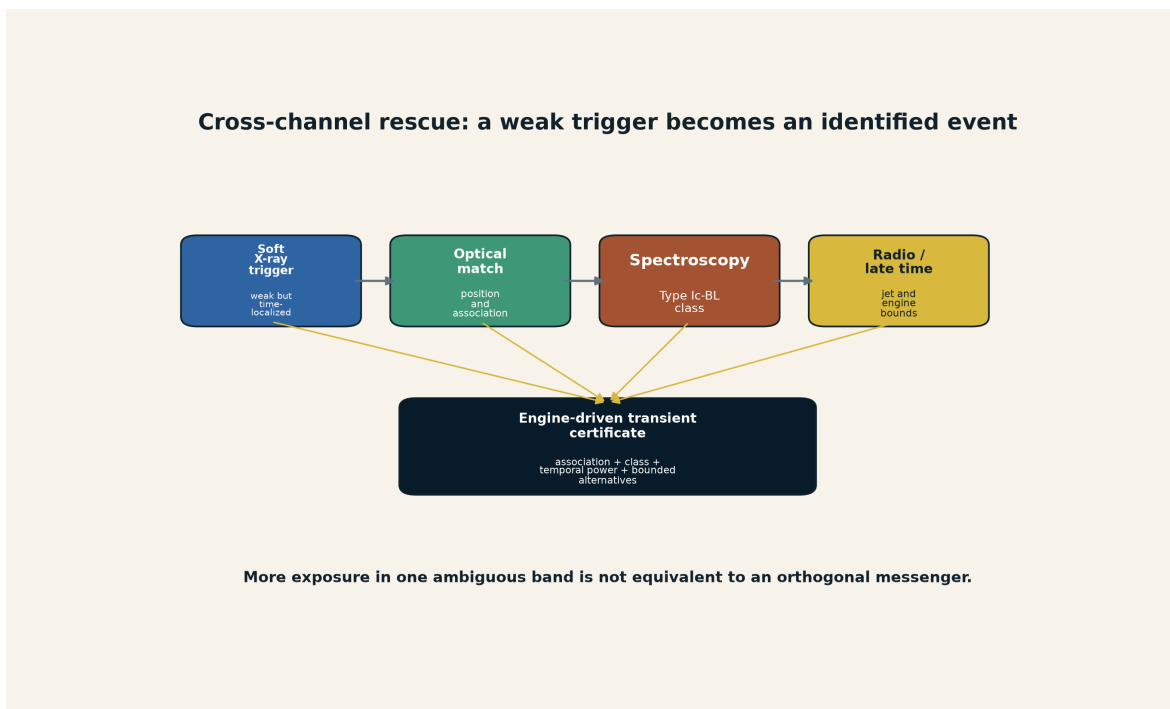


Figure 14: [MODEL] Cross-channel rescue. A weak soft-X-ray trigger becomes a classified engine-driven supernova only after optical association, spectroscopy, temporal evolution, and radio constraints. The diagram does not imply that every X-ray flash shares one engine. Creator: ASTRA / Jacko T. Source: original synthesis based on Srinivasaragavan et al. License: CC BY 4.0.

19.3 Category audit

The observed class can be written schematically as

$$\text{category} = f(\text{event, band, threshold, cadence, follow-up, forward model}).$$

This does not make categories arbitrary. It makes their selection functions part of the scientific record. A soft-X-ray mission can reveal engine-driven explosions that historical gamma-ray triggers would undercount or classify differently.

Status: X-ray flash/supernova association **Established**; sustained central engine **Strong inference**; exact magnetar-disk-CSM model **Plausible**; universal engine for the class **Unsupported**.

20. Dark-photon resonance and self-detuning media

20.1 The linear exclusion argument

Kinetically mixed dark-photon dark matter behaves in an ionized medium like a weak oscillating electric drive. When its mass matches the plasma frequency,

$$m_{A'} \simeq \omega_p, \quad \omega_p^2 = \frac{4\pi n_e e^2}{m_e},$$

The displayed 4π form uses Gaussian-cgs electromagnetic units. In SI units

the corresponding relation is $\omega_p^2 = n_e e^2 / (m_e \varepsilon_0)$. A calculation must fix one unit system rather than mix these forms.

Linear treatments predict resonant transfer into a nearly zero-wavenumber Langmuir mode. Previous cosmological limits extrapolated that transfer into enough plasma heating to affect ionization history or the cosmic microwave background.

20.2 Nonlinear saturation

Hook, Huang and Shalaby report particle-in-cell simulations in which the driven Langmuir wave reaches roughly the electron thermal-energy scale. The ponderomotive force then excites finite-wavenumber Langmuir and ion-acoustic modes, creates strong density inhomogeneity, shifts the local plasma frequency, and suppresses further coherent conversion. The resulting energy deposition is orders of magnitude below the linear cosmological threshold. The reported bounds weaken by factors of approximately 3,000 to 10^7 across ten orders of magnitude in dark-photon mass. (Hook, Huang, and Shalaby 2026)

The causal loop is

$$A' \rightarrow E_{\text{plasma}} \rightarrow \delta n_e \rightarrow \delta \omega_p \rightarrow \text{detuning} \rightarrow \text{saturation}.$$

20.3 Scope and unresolved issues

The simulations do not reproduce the full hierarchy between plasma and cosmological timescales. Collisions, expansion, ionization history, multidimensional turbulence, and long-time energy transport require further work. The

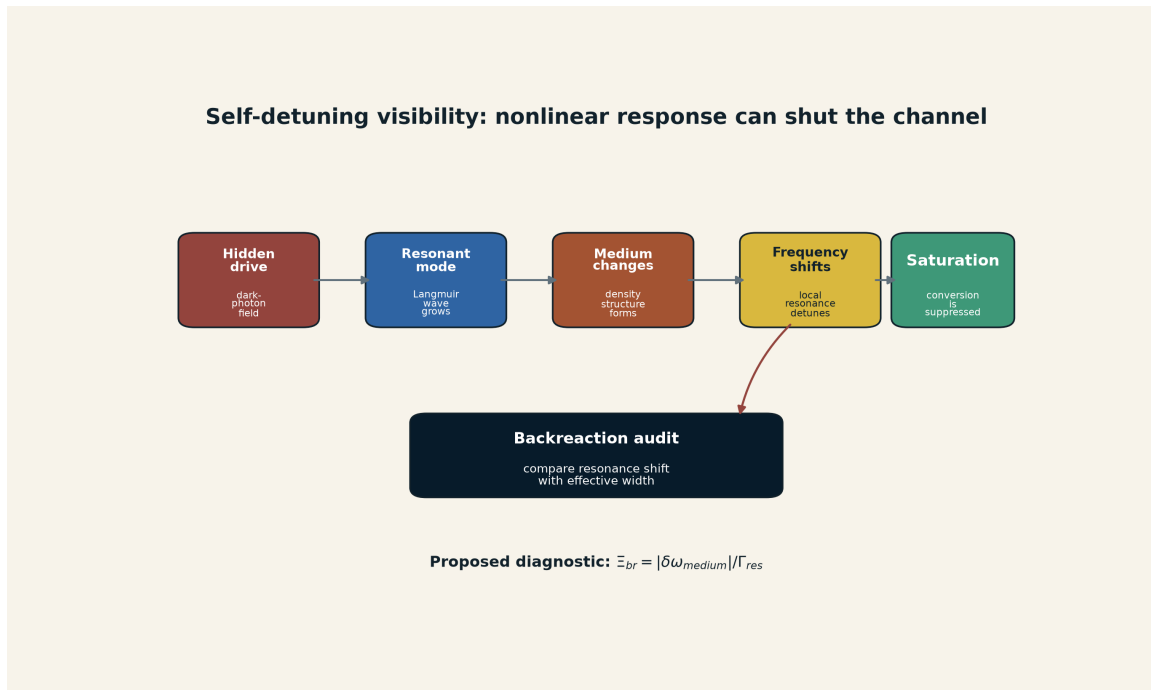


Figure 15: [MODEL] Self-detuning resonant visibility. A hidden drive excites a plasma mode; the growing mode reorganizes the plasma density, shifts the resonance, and suppresses further conversion. The diagram summarizes the reported mechanism but is not a particle-in-cell result. Creator: ASTRA / Jacko T. Source: original synthesis based on Hook et al. License: CC BY 4.0.

correct conclusion is therefore narrower than the paper’s title taken literally: 1223
the existing resonant constraints appear to rely on a self-inconsistent linear 1224
extrapolation, and the parameter space is substantially reopened under the 1225
reported nonlinear model. 1226

This is not evidence that dark photons exist. It is a demonstration that a null 1227
constraint can fail when the transducer backreacts before the assumed signal 1228
accumulates. 1229

Status: nonlinear saturation in the implemented simulations **Established** 1230
within that declared model; substantial weakening of previous linear bounds 1231
Strong inference; final cosmological exclusion map **Open**; dark-photon iden- 1232
tity **Open**. 1233

21. Neptune and catastrophic tomography 1234

21.1 Observation 1235

JWST spectra of Larissa, Galatea, Proteus, and Neptune’s rings show deep 1236
absorption near $3\ \mu\text{m}$ but no exposed water-ice bands. Larissa, Galatea, and 1237
the rings additionally show a sharp $2.72\ \mu\text{m}$ feature diagnostic of magnesium- 1238
rich phyllosilicates, implying extensive past water-rock alteration. The cur- 1239
rent small moons are too small to plausibly generate that degree of internal 1240
alteration in their present form. ([Davis et al. 2026](#)) 1241

The favored interpretation is that the material came from the deep interiors 1242
of larger primordial satellites destroyed during Triton’s violent capture and 1243
orbital evolution, or from a tidally shredded differentiated dwarf planet. The 1244
present moons and rings then reaccreted from selected debris. A companion 1245
Science Advances study argues that Nereid’s composition and dynamics are 1246

1247 more consistent with an original regular satellite scattered by Triton's cap-
 1248 ture than with an ordinary captured Kuiper Belt object. (Davis et al. 2026;
 1249 Belyakov et al. 2026)

1250 21.2 Catastrophe as an archive operator

1251 Represent the present body as

$$X_{\text{present}} = \mathcal{R}_{\text{reaccretion}} \circ \mathcal{C}_{\text{catastrophe}}(X_{\text{parent}}).$$

1252 The catastrophe can expose a parent interior while destroying depth, parent
 1253 identity, chronology, and original geometry. Information gain and information
 1254 loss occur together.

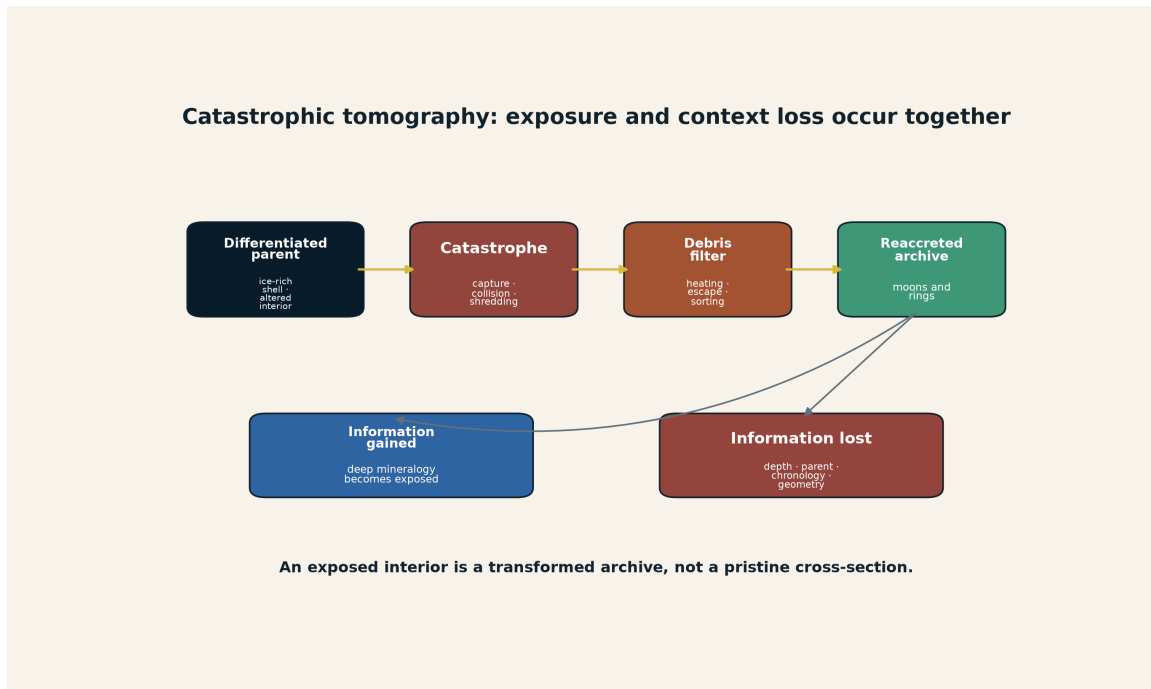


Figure 16: [MODEL] Catastrophic tomography in the Neptune system. Destruction can expose aqueously altered interior material while fragmentation, volatile loss, sorting, and reaccretion erase original context. The diagram is a causal audit, not a unique reconstruction of Triton's capture. Creator: ASTRA / Jacko T. Source: original synthesis based on Davis et al. and Belyakov et al. License: CC BY 4.0.

The absence of exposed water ice remains a major unresolved constraint. Any 1255
successful history must explain how altered rocky interior material was pref- 1256
erentially retained or exposed while water-rich outer material escaped, was 1257
buried, was altered, or became spectrally masked. 1258

Status: hydrated/phylosilicate spectral features **Established**; altered larger 1259
parent bodies **Strong inference**; destruction during Triton capture **Strong** 1260
inference; exact collision/reaccretion sequence and missing-ice mechanism 1261
Open. 1262

22. Adjacent breaking-physics calibrations 1263

Three adjacent records reinforce the operator-aware program without enter- 1264
ing the physical core as validated laws. 1265

A proposed lunar-orbiting CosmoCube instrument would use the Moon's far- 1266
side as a radio shield to observe the 10-50 MHz redshifted 21-cm signal from 1267
the cosmic dark ages. The Moon becomes part of the observation boundary. 1268
The mission remains under development; no dark-ages detection is claimed. 1269
([CosmoCube Collaboration 2026](#)) 1270

A model of the Cygnus Bubble proposes that ultra-high-energy cosmic rays es- 1271
caped from Cygnus X-3 and generated a spatially displaced gamma-ray halo. 1272
The emission site would therefore not coincide with the acceleration site. The 1273
model is accepted for publication and prospectively testable through morphol- 1274
ogy and energy dependence, but is not yet uniquely established. ([Shi et al.](#) 1275
[2026](#)) 1276

An accepted theoretical study of strongly magnetized active-galactic-nucleus 1277
tori finds conditions under which streaming instability and rapid accretion 1278

1279 could produce enormous populations of planet-mass and even stellar-mass ob-
1280 jects. No planet in an AGN torus has been observed; the result is a formation-
1281 channel prediction, not a census. ([Mishra et al. 2026](#))

1282 Together these cases reinforce a disciplined rule: natural bodies can act as
1283 shields, converters, transport media, or generative architectures, but their
1284 role must be specified by a forward model and a prospective observation.

1285 **Part VI - Scientific applications and limits**

1286 **23. Planetary phase-reservoir applications**

1287 **23.1 What changes in SPPT**

1288 The original SPPT proposition remains that planetary state depends on phase-
1289 reservoir connectivity. The v1.0.8 candidate adds that the edges themselves
1290 may contain history-dependent constitutive state. A more complete planetary
1291 representation is

$$\mathcal{P}^*(t) = (\mathcal{G}(t), x(t), \theta, u(t), b_E(t)).$$

1292 This extension matters when an interface possesses memory on a timescale
1293 comparable with the process of interest. Candidate examples include:

- 1294 • fault or subduction interfaces with damage, mineral reactions, pore pres-
1295 sure, and permeability history;
- 1296 • phase boundaries with nucleation barriers and wetting history;
- 1297 • compositional gradients that change conductivity or convection;
- 1298 • core-mantle boundary layers with evolving morphology and chemical ex-

change;

1299

- cloud, haze, or condensate interfaces with particle-size and charge history; 1300
- ocean-ice or rock-water boundaries with salinity, porosity, fracture, and re- 1301
active transport; 1302
- magnetic or plasma boundaries whose coupling depends on field orientation, 1303
reconnection state, or rotating geometry. 1304

The extension does not imply that every interface requires new state variables. 1305

It supplies a promotion test. 1306

23.2 Candidate planetary edge record 1307

For a planetary edge e , record 1308

$$b_e = (T, P, \phi, k, D, M, C_f, \gamma, \kappa, \sigma, \mathbf{B}, \mathcal{G}_e),$$

where ϕ is porosity, k permeability, D damage, M phase/mineral assemblage, 1309
 C_f fluid composition, γ interfacial energy, κ thermal or electrical conductivity, 1310
 σ stress or charge state, $\mathbf{B}$ magnetic field, and $\mathcal{G}_e$ local geometry. Not every 1311
application uses every coordinate. 1312

The edge flux can be written schematically as 1313

$$J_e = G_e(\Delta\tilde{\mu}, \Delta T, \Delta P, \Delta\Phi, b_e, \mu_e, \mathcal{A}_e).$$

This is a template. A real model must replace G_e with a validated law, phase 1314
diagram, transport closure, or numerical solver. 1315

1316 **23.3 Nonreciprocity in planetary models**

1317 A planetary system may contain effective directionality without violating
1318 global conservation. Gravitational settling, irreversible reactions, radiative
1319 escape, chemically mediated transport, and rotating magnetized flows can
1320 produce directed reduced couplings. The correct question is not whether a
1321 matrix is symmetric by default. It is:

- 1322 1. what variables were eliminated;
- 1323 2. what reservoir supplies the free energy;
- 1324 3. which momentum or species flux crosses the model boundary;
- 1325 4. whether the entropy ledger is nonnegative;
- 1326 5. whether the asymmetry survives coarse-graining;
- 1327 6. whether it changes held-out observables.

1328 The nonreciprocal-colloid experiment motivates this audit. It does not estab-
1329 lish a nonreciprocal law for any planet.

1330 **23.4 Stateful interfaces and planetary hysteresis**

1331 The catalyst provides a compact analogy for a planetary interface that
1332 changes under operation. Examples include a fault that fractures and later
1333 seals, a magma-ocean boundary that crystallizes and partitions species, an
1334 atmosphere-surface interface that oxidizes the surface and changes future
1335 uptake, or an ice-shell fracture that changes permeability and heat transfer.

1336 The useful common form is

$$J_e = G_e(X, b_e), \quad \dot{b}_e = F_e(X, b_e, J_e),$$

not a claim that the same microscopic law applies. 1337

23.5 Observation design 1338

The existing synthetic supplement showed that static surface equilibrium can 1339
conceal internal topology and that multi-frequency complex response can re- 1340
duce a capacity-conductance degeneracy. (Jacko T. 2026i) 1341

The v1.0.7 program broadens the intervention set: 1342

- different forcing frequencies and amplitudes; 1343
- mode reversal; 1344
- several observation ports; 1345
- transient relaxation after forcing removal; 1346
- controlled or naturally occurring support migration; 1347
- comparison across objects with different boundary histories; 1348
- sector measurements that close environment exchange; 1349
- population-level variation in which the same edge law is tested under differ- 1350
ent states. 1351

The fundamental criterion remains held-out prediction against simpler fixed- 1352
topology and memoryless-edge baselines. 1353

24. Earth, origins, and boundary-state science 1354

24.1 Relationship to *Earth Is the Instrument* 1355

The Earth supplement develops a wider historical and biological inference 1356
framework. Its useful contribution to v1.0.7 is not a new planetary equation. 1357
It is a disciplined separation of: 1358

- 1359 • physical state;
- 1360 • boundary or interface state;
- 1361 • control;
- 1362 • observation;
- 1363 • archive;
- 1364 • candidate generators;
- 1365 • certificate scope.

1366 The original Earth working paper states that geology can preserve, transform,
 1367 expose, and censor evidence; its page-17 funnel diagram presents observed
 1368 history as the result of deposition, alteration, exposure, recognition, and inter-
 1369 pretation rather than a neutral movie. It also presents a boundary-state ladder
 1370 from planetary differentiation through living membranes, external memory,
 1371 scientific instrumentation, and spacefaring biospheric capability. Those are
 1372 conceptual diagrams, not demonstrations of intention or one hidden machine.
 1373 The foundational thesis is retained in its own line. ([Jacko T. 2026f](#), [2026a](#))

1374 **24.2 Distributed geological nursery**

1375 A stateful-edge SPPT description strengthens the distributed-origin model
 1376 without proving it. Prebiotic chemistry may have depended on transport
 1377 among environments with different active supports and edge states:

$$\mathcal{N}_{\text{origin}} = (E_i, T_{ij}, \mathcal{R}_i, b_{ij}, u(t)).$$

1378 A shoreline, mineral pore, hydrothermal interface, sediment aquifer, aerosol,
 1379 or wet-dry pool may each perform a different operation. The edge variables
 1380 may include wetting history, mineral surface composition, redox state, salinity,

temperature cycling, permeability, and adsorption coverage.

1381

The proposed moving-front coordinate

1382

$$\Xi = v_f \tau_{\text{int}} / \ell_{\text{int}}$$

suggests a specific experimental program: move a UV, pH, redox, thermal, or hydration front across a mineral network while holding total energy and chemical inventory fixed. Measure whether reaction yield depends on the match among front speed, intermediate lifetime, and interaction length. A peak near $\Xi \sim 1$ would be new evidence for mode-resolved coupling; its absence would demote the proposal.

1383

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24.3 Biological and ecological nonreciprocity

1389

Living systems routinely exchange matter and energy with environments and can exhibit effective asymmetric interactions. The colloid work provides a controlled physical model for how asymmetry can sustain dynamic clusters. It does not show that biological collectives use the same electrohydrodynamic mechanism.

1390

1391

1392

1393

1394

A biological application would need:

1395

- a specific mediator such as chemical signal, fluid flow, mechanical force, or sensory response;
- directional coupling coefficients;
- energy source and dissipation;
- perturbations that reverse or symmetrize the interaction;
- measurements across scales to test whether nonreciprocity survives coarse-graining;

1396

1397

1398

1399

1400

1401

1402

1403 • a residue or function that differs from reciprocal controls.

1404 **24.4 Human origins and the archive**

1405 The candidate does not materially change ASTRA's existing origin ledger.
1406 Terrestrial biological nesting, Earth-life coevolution, exogenous chemical
1407 input, and a substantially missing coastal/perishable human archive remain
1408 supported at their existing levels. Natural panspermia remains open; directed
1409 early seeding remains unsupported; recent global industrial predecessors
1410 remain strongly constrained by expected cross-archive residues. (Jacko T.
1411 2026a)

1412 The new methods improve the questions asked of missing evidence:

- 1413 • Which physical sector carried the trace?
- 1414 • Which boundary transformed it?
- 1415 • Where was the active support?
- 1416 • Did a sampling operator systematically exclude it?
- 1417 • Which environment flux would be required to sustain the proposed process?
- 1418 • Is the absence local to one observable sector or repeated across indepen-
1419 dent high-visibility channels?

1420 An omitted sector is not permission to insert any preferred history. It be-
1421 comes a scientific possibility only when the sector, transformation, visibility,
1422 and prospective observation are specified.

25. Cosmology and dark matter

1423

25.1 Hidden state remains compound

1424

The exact-commit Cosmic Visibility draft represents a dark-matter hypothesis 1425
as more than a particle mass, separating physical and phase-space state, gen- 1426
esis and abundance history, coupling, and formation history from the visibility 1427
model: 1428

$$\mathcal{H}_{\text{DM}} = (S_{\chi}, G_{\chi}, C_{\chi}, f_{\chi}, H_{\chi}),$$

where S_{χ} is physical state, G_{χ} genesis, C_{χ} coupling, f_{χ} phase-space distribu- 1429
tion, and H_{χ} formation and phase history. This tuple is candidate bookkeep- 1430
ing synthesized from that inspectable repository source; it is not a claim that 1431
one detector directly measures five uniquely separable coordinates. (Jacko T. 1432
2026e) 1433

The v1.0.8 candidate adds two cautions. 1434

First, a proposed hidden-sector interaction must close its energy, momentum, 1435
and abundance ledgers. Effective nonreciprocity or sector conversion does 1436
not remove the need for a Lagrangian or effective operator. 1437

Second, a null result constrains a source-visibility combination. Detector ba- 1438
sis, local halo model, mediator, coherence time, and backgrounds are part of 1439
the certificate. 1440

25.2 Required dark-sector adapter

1441

A physical bridge requires an interaction such as 1442

$$H_{\text{int}} = \sum_a g_a O_a^{\text{SM}} \otimes O_a^{\chi},$$

1443 plus:

- 1444 • state or phase-space distribution;
- 1445 • mediator and coupling normalization;
- 1446 • production and abundance history;
- 1447 • mass and coherence range;
- 1448 • detector transfer function and units;
- 1449 • environmental nuisance model;
- 1450 • predictions in more than one observable sector;
- 1451 • preregistration or blind injection;
- 1452 • a null region that would materially demote the model.

1453 Without that contract, “the detector used the wrong observable” is a logical
 1454 possibility rather than an explanation.

1455 **25.3 Cosmic filaments as natural transducers**

1456 The cosmic-filament dark-decay work proposes that a hidden carrier can be
 1457 converted into photons in magnetized large-scale structure. The filament is
 1458 therefore part of the apparatus, not merely background. ([Dunsky, Krnjaic, and](#)
 1459 [Pinetti 2026; Jacko T. 2026e](#))

1460 The expected signal depends on source abundance and lifetime, branching
 1461 fraction, field strength, coherence length, filament geometry, propagation,
 1462 and gamma-ray response. A null maps a compound parameter manifold. The
 1463 highest-information next test is spatial: correlate residual gamma-ray maps
 1464 with independently reconstructed filament, lensing, Faraday-rotation, and

baryon tracers, and compare with matched void controls. 1465

No such analysis presently identifies dark matter or gravitons. The valid 1466
bridge is operator-aware cosmology. 1467

25.4 Primordial memory 1468

The galaxy-spin work suggests that present angular momentum can retain 1469
information about primordial tidal fields acting on protohalos. (Sheng et al. 1470
2026) 1471

ASTRA treats that as a residue channel: 1472

primordial field $\rightarrow$ dark-matter halo history $\rightarrow$ galaxy spin $\rightarrow$ observed kinematics.

The next step is a transfer function across mass, redshift, scale, baryonic 1473
tracer, and dark-matter model. A correlation is not a particle identification. It 1474
is a new archive of gravitational history. 1475

25.5 Neutrino fog and sector completeness 1476

XENONnT's low-threshold null constrains specified masses, couplings, halo 1477
assumptions, detector response, and backgrounds. It approaches a regime 1478
where solar-neutrino and light-dark-matter recoils become observationally 1479
similar. (XENON Collaboration 2026) 1480

More exposure in one target may provide diminishing epistemic rent. Differ- 1481
ent nuclei, electron channels, phonons, magnons, timing, directionality, and 1482
astrophysical measurements supply different observation sectors. The cor- 1483

1484 rect portfolio objective is not maximum count alone. It is maximum generator
1485 separation under bounded cost and risk.

1486 **26. Mathematics as a calibration of certificate** 1487 **scope**

1488 **26.1 The Jacobian lesson**

1489 A July 2026 preprint claims explicit counterexamples to the Jacobian conjec-
1490 ture in complex dimension three and higher, while the two-variable case re-
1491 mains open. This candidate has checked the bibliographic preprint record but
1492 has not independently validated the claimed counterexamples, so it uses the
1493 paper only as a provisional local-to-global calibration. ([Gao 2026](#))

1494 The lesson for ASTRA is exact and limited:

$$\det DF \equiv c \neq 0$$

1495 certifies local nonsingularity, not global injectivity. The higher-dimensional
1496 counterexamples fail through nonproper behavior at infinity. A true local cer-
1497 tificate was promoted too far.

1498 The v1.0.7 local-to-global stack therefore asks:

- 1499 1. What local response was certified?
- 1500 2. What complete fibers or generator classes remain?
- 1501 3. Which boundary or escape direction was omitted?
- 1502 4. Which arithmetic reductions or implementation checks apply?
- 1503 5. What exact proof object or collision witness closes the claim?

The mathematical case does not imply that the universe is simulated or holo- 1504
 graphic. It gives a rigorous toy example of the methodological warning that 1505
 local reversibility does not ensure global uniqueness. 1506

26.2 Prime reductions 1507

For an integer polynomial map, reduction modulo prime p provides a family of 1508
 arithmetic observation channels. Splitting and collision patterns can reveal 1509
 hidden branch structure. But a modular collision does not automatically lift to 1510
 a characteristic-zero counterexample, and absence of collisions over $\mathbb{F}_p$ does 1511
 not exclude collisions over extensions or at infinity. 1512

The correct use is diagnostic and certificate-aware: 1513

local Jacobian $\rightarrow$ fiber/collision scheme
 $\rightarrow$ boundary at infinity $\rightarrow$ prime spectrum
 $\rightarrow$ exact lifting or obstruction.

This remains a separately scoped arithmetic research program, not a plane- 1514
 tary model. 1515

26.3 Simulation and holography 1516

Simulation hypotheses and holographic dualities concern representation, en- 1517
 coding, and observer access. The Jacobian calibration shows only that a lo- 1518
 cally nondegenerate map can still fail globally if state escapes the modeled 1519
 boundary. 1520

A specific simulation model becomes scientific only if it predicts residues such 1521
 as finite precision, anisotropy, resource cutoffs, nonunitarity, or external in- 1522

interventions. A holographic model requires a defined bulk-boundary dictionary, code subspace, observables, and reconstruction limits. No result in SPPT/ASTRA proves or disproves either broad idea.

27. One methodology, domain-specific laws

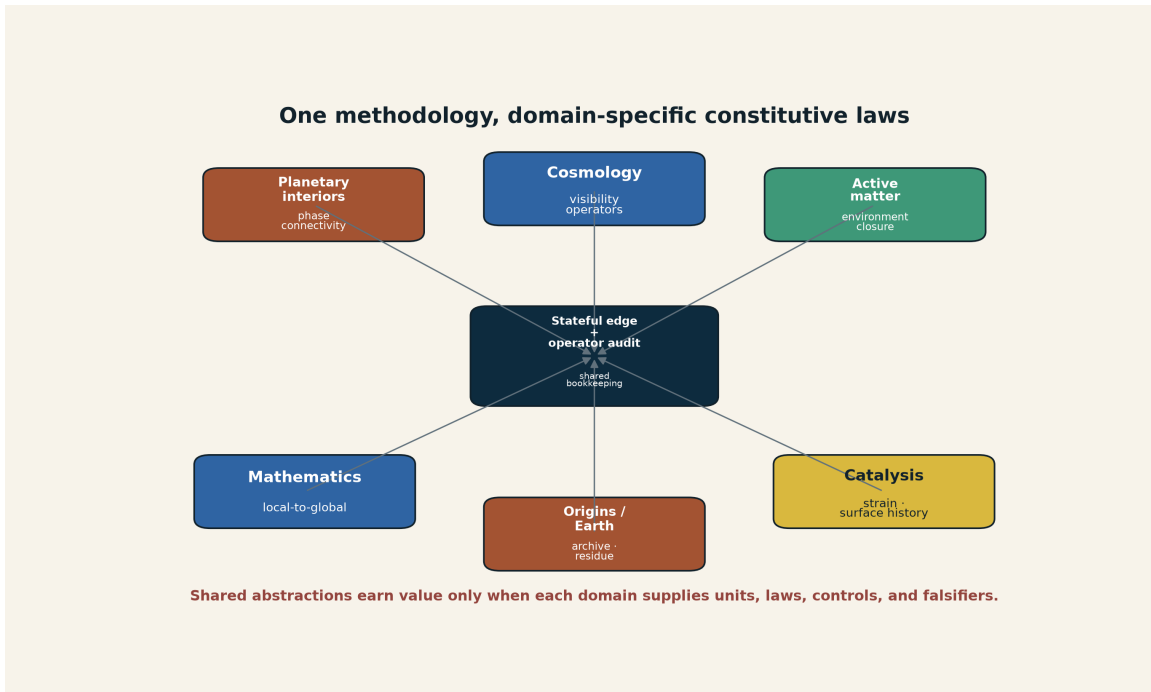


Figure 17: [MODEL] Application map. Stateful-edge and operator-aware audits provide common bookkeeping across domains, while each domain retains its own constitutive laws, units, and falsifiers. Creator: ASTRA / Jacko T. Source: original vector model. License: CC BY 4.0.

The candidate’s strongest unifying statement is methodological:

The observed future is determined by node states, graph connectivity, edge state, active support, environment exchange, and observation operators. Their relative importance is query- and scale-dependent.

That statement does not erase the differences among electrodynamics,

electrocatalysis, planetary thermodynamics, quantum measurement, cosmology, archaeology, and algebraic geometry. 1533
1534

Part VII - New benchmarks, claim admission, and release engineering 1535 1536

28. Proposed v1.0.8 benchmark suite 1537

28.1 Benchmark A: nonreciprocal pair closure 1538

Implement the exact two-particle reduced model with attraction and short-range repulsion. Verify: 1539
1540

- pair separation converges to a declared r_* ; 1541
- center velocity equals $(a_{LS} - a_{SL})r_*/2$; 1542
- velocity vanishes when couplings are reciprocal; 1543
- the particle momentum residual equals the declared environment exchange 1544
in the enlarged bookkeeping model; 1545
- sign reversal of asymmetry reverses pair motion; 1546
- dimensional checks pass. 1547

This is an analytic/synthetic benchmark, not a fit to Hara et al. 1548

28.2 Benchmark B: dynamic arrest versus frozen arrest 1549

Create a minimal cluster process with aggregation and asymmetry-dependent fragmentation. Compare: 1550
1551

- reciprocal coarsening; 1552
- frozen kinetic arrest; 1553

- 1554 • dynamic arrested coarsening;
- 1555 • out-of-set correlated-noise or heterogeneity controls.

1556 Report cluster scale, turnover, edge lifetime, motif flux, and entropy or dis-
 1557 sipation proxies. Require a classifier to distinguish frozen from dynamically
 1558 renewed states.

1559 The generating model should be omitted from some candidate sets to test
 1560 family rejection.

1561 **28.3 Benchmark C: self-rewriting catalyst edge**

1562 Construct a two-timescale edge model:

$$\dot{\varepsilon} = -\frac{\varepsilon - \varepsilon_u(t)}{\tau_\varepsilon}, \quad \dot{c}_{\text{surf}} = F_c(c_{\text{surf}}, j, T),$$

$$j = G(\eta, \varepsilon, c_{\text{surf}}).$$

1563 Synthetic protocols should include:

- 1564 • strain-only calibration;
- 1565 • dealloying-only calibration;
- 1566 • cross interaction;
- 1567 • cycling reversal;
- 1568 • withheld long-duration regime;
- 1569 • model mismatch where roughness, not composition, is the omitted genera-
 1570 tor.

1571 The objective is to test identifiability, not to simulate the real catalyst in detail.

28.4 Benchmark D: active-support omission

1572

Generate a spatiotemporal response with a known support kernel. Fit candidates that: 1573
1574

- recover the correct support family; 1575
- use a shifted support; 1576
- use a broad uniform support; 1577
- omit support and alter bulk parameters; 1578
- omit the true support from the candidate set. 1579

Evaluate held-out interventions that move the support while preserving total input. This is the decisive test of whether active support pays predictive rent. 1580
1581

28.5 Benchmark E: sector and visibility composition

1582

Combine the four-generator Sector-Complete benchmark with a visibility kernel and detector confusion. Require the analysis to distinguish: 1583
1584

- source absence; 1585
- sector migration; 1586
- propagation loss; 1587
- detector inefficiency; 1588
- out-of-set hybrid behavior. 1589

Report the observational quotient, Fisher null directions, mutual information with prior sensitivity, and family adequacy. 1590
1591

28.6 Benchmark F: endogenous visibility and self-detuning

1592

Construct four candidate generators under the same terminal observation channel: 1593
1594

- 1595 1. a fixed linear transducer;
- 1596 2. a source-coupled self-detuning transducer;
- 1597 3. a scattering envelope that changes line shape without changing the source
- 1598 state;
- 1599 4. a catastrophic-remixing archive that exposes interior composition while
- 1600 randomizing provenance.

1601 The local terminal record should leave at least two models observationally
1602 equivalent. Add an independent measurement of transducer state, an altered
1603 forcing protocol, or a second messenger and test whether the quotient refines.
1604 Include out-of-set hybrids and reject the candidate family when none fits held-
1605 out data.

1606 The benchmark is synthetic. Success would validate implementation and
1607 experimental-design logic only; it would not validate black-hole envelopes,
1608 dark photons, supernova engines, or Neptune's history.

1609 **28.7 Benchmark G: bridge-protocol end-to-end replay**

1610 Run every candidate through:

- 1611 1. conservation validation;
- 1612 2. thermodynamic ledger;
- 1613 3. finite equivalence diagnostics;
- 1614 4. intervention selection;
- 1615 5. calibration/test split;
- 1616 6. held-out scoring;
- 1617 7. out-of-set rejection;
- 1618 8. hash and environment recording.

The benchmark should preserve negative outcomes. A model that wins a section score but loses held-out prediction must not be rewritten as successful topology recovery.

29. Proposed claim-admission structure

29.1 Tier A - retained core claims

All 55 v1.0.7 consequential claims remain in place with their exact public IDs, statements, support, limitations, evidence classes, and dispositions. The candidate package embeds the v1.0.7 CLAIM_MATRIX.json byte-for-byte and deterministically projects it into the candidate ledger before adding any V108 claim.

29.2 Tier B - new exact or definitional claims

Candidate claims suitable for hand or mechanical verification include:

- the symmetric/antisymmetric decomposition of a finite coupling matrix;
- the two-particle pair-drift identity under the declared toy equations;
- the environment closure identity for a partitioned momentum ledger;
- the observational-equivalence relation and quotient;
- the corrected channel/POVM measurement equations;
- the active-support gauge transformation;
- the distinction among fixed topology, stateful edge, and topology change;
- dimensional validity of Ξ ;
- the self-rewriting edge model as proposed syntax.

Only the first eight can be admitted as exact mathematical or implementation

1641 statements under their definitions. The self-rewriting edge is a framework
1642 proposal.

1643 **29.3 Tier C - external calibration claims**

1644 The Hara and Redondo results should enter as source-local claims with exact
1645 numerical and scope boundaries. They should not be used to validate SPPT or
1646 ASTRA globally. Every sentence should identify whether it is an observation,
1647 authors' interpretation, or ASTRA structural inference.

1648 **29.4 Tier D - proposed scientific applications**

1649 Planetary, origin-of-life, biological, dark-matter, and cosmological applications
1650 remain proposed_only until a domain-specific model and data test exist. The
1651 project should not promote them because the methods are coherent or visually
1652 compelling.

1653 **30. Source, rights, and provenance policy**

1654 **30.1 News as discovery lead**

1655 Phys.org and similar reports are discovery leads. The current update follows
1656 the repository's production rule: technical claims are grounded in primary
1657 papers or stable official records. The news pages are not copied into the pack-
1658 age, and their images are not reused.

30.2 Redrawn figures

1659

The ORR comparison chart is an original redraw of three reported numbers. 1660
It is not a reproduction of the paper’s graphical abstract or figures. The non- 1661
reciprocal and catalyst schematics are original models and are labeled accord- 1662
ingly. 1663

30.3 AI assistance and responsibility

1664

AI assistance may organize sources, draft equations, generate code, and per- 1665
form adversarial review. It is not an author, rights holder, peer reviewer, or 1666
scientific validator. The human maintainer remains responsible for source 1667
selection, wording, release decisions, and any claim of originality. 1668

30.4 Repository identity

1669

A candidate document must record: 1670

- base commit and tree; 1671
- exact admitted file hashes; 1672
- source versions and retrieval date; 1673
- generated output hashes; 1674
- build environment and dependencies; 1675
- author and license map; 1676
- whether any file is inherited from an immutable release; 1677
- whether the candidate changes a core claim. 1678

No candidate should reuse the v1.0.6 or v1.0.7 release identity or manifest. 1679
A v1.0.8 release must have its own tag, assets, source archive, identity JSON, 1680
and remote read-back verification. 1681

31. Fail-closed promotion gates

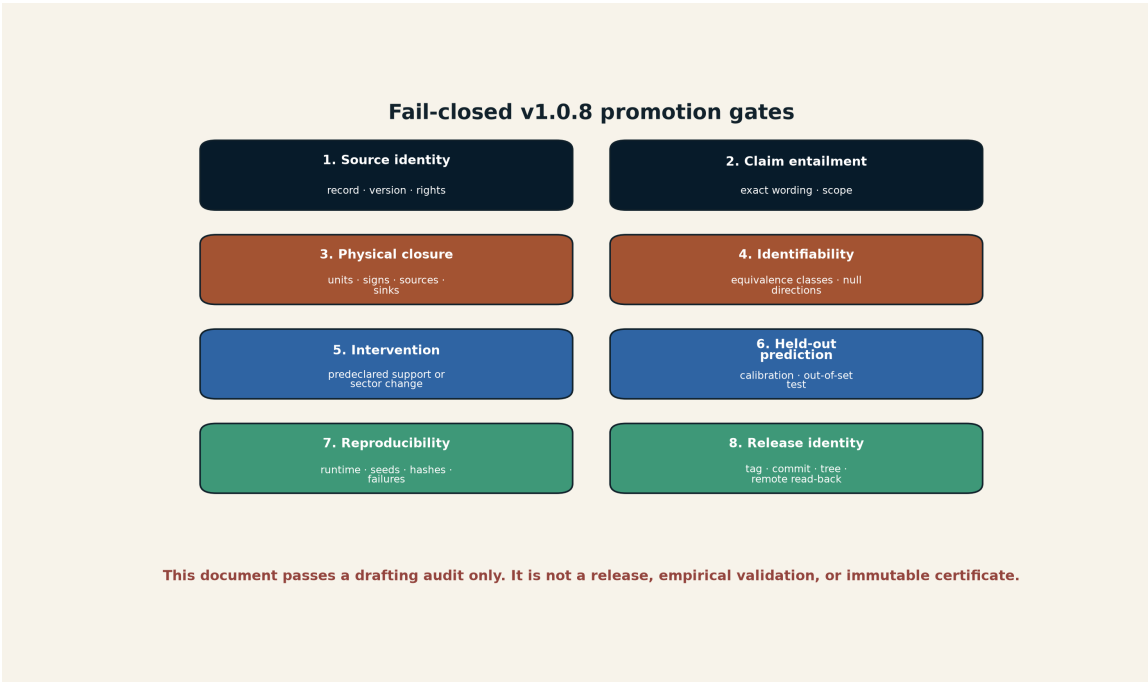


Figure 18: [MODEL] Proposed promotion gates for a real v1.0.8 release. This document has completed only the drafting and source-integration stage. Creator: ASTRA / Jacko T. Source: original vector model. License: CC BY 4.0.

A release candidate should fail if any of the following occurs:

1. a new claim lacks a primary source or exact code/equation locator;
2. a source does not support the exact wording;
3. a directed edge lacks an environment closure record;
4. a support kernel has hidden units or an unacknowledged normalization gauge;
5. an observation null is generalized beyond its sectors;
6. a model family cannot reject an out-of-set generator;
7. calibration and test data leak;
8. a negative held-out result is omitted;
9. a supplemental analogy is presented as planetary evidence;
10. the new tree cannot be rebuilt in two independent environments;

11. PDF/DOCX accessibility and visual preflight fail; 1695
12. release assets, tag, identity JSON, and remote bytes do not reconcile. 1696

32. Candidate release plan 1697

32.1 Stage 0 - preserve v1.0.7 1698

Do not edit or retag v1.0.7 (or the historical v1.0.6 release). Embed the v1.0.7 1699
claim matrix byte-for-byte in a new candidate namespace and verify its hash 1700
and projected claims against the immutable tag. 1701

32.2 Stage 1 - integrate schemas and contracts 1702

Create a new `stateful_edge.schema.json` or extend the existing bridge pro- 1703
tocol with: 1704

- edge state; 1705
- mode and active support; 1706
- reciprocity class; 1707
- environment exchange ledger; 1708
- observation sectors; 1709
- visibility operator; 1710
- equivalence class; 1711
- falsifier; 1712
- evidence disposition. 1713

The schema dialect must be validated by the declared validator. An 1714
environment-limited result is preferable to a false pass. 1715

1716 **32.3 Stage 2 - implement benchmarks**

1717 Add the seven proposed synthetic benchmarks with fixed seeds, exact ex-
1718 pected outputs where possible, and omitted-generator controls. Ensure that
1719 candidate selection, family adequacy, and held-out prediction are separate
1720 outputs.

1721 **32.4 Stage 3 - manuscript integration**

1722 Rewrite the core manuscript rather than appending a loose supplement.
1723 Preserve the existing exact derivations, then add stateful edges, closure-
1724 conditioned nonreciprocity, active support, sector completeness, visibility,
1725 and bridge protocol in one notation system.

1726 The Earth and human-origin material should remain a separately cited sup-
1727 plemental line. The main planetary paper should summarize its transferable
1728 methods without absorbing its full historical content.

1729 **32.5 Stage 4 - claim and source audit**

1730 Generate the claim matrix and source ledger from one frozen tree. For every
1731 new claim record:

- 1732 • exact sentence or equation locator;
- 1733 • source or code locator;
- 1734 • hypotheses, units, domain, and quantifiers;
- 1735 • evidence class;
- 1736 • limitation and counterexample;
- 1737 • admission disposition;
- 1738 • test or reproduction command;

- output hash. 1739

32.6 Stage 5 - release engineering 1740

Build canonical PDF, HTML, technical supplement, source archive, check- 1741
sums, and release identity. Run: 1742

- unit and integration tests; 1743
- deterministic data reproduction; 1744
- claim-source coverage checks; 1745
- manifest replay; 1746
- dependency and license audit; 1747
- PDF structure, font, link, and renderer preflight; 1748
- DOCX accessibility if a reading edition is released; 1749
- clean clone build; 1750
- natural main workflow; 1751
- tag workflow; 1752
- remote asset read-back. 1753

Only after those gates pass should a new v1.0.8 tag or release be considered. 1754

33. Consolidated research program 1755

The next scientific work should prioritize experiments that separate mecha- 1756
nisms rather than expand vocabulary. 1757

33.1 Active-matter program 1758

- Reproduce the bidisperse colloid result independently. 1759
- Measure mediator flow and environment momentum exchange. 1760

- 1761 • Map nonreciprocity across particle-size ratio, field, density, and viscosity.
- 1762 • Test whether nonreciprocity survives coarse-graining.
- 1763 • Develop a motif-transition and dynamic-arrest benchmark.

1764 **33.2 Mechanochemical catalyst program**

- 1765 • Separate strain from dealloying and active area.
- 1766 • Measure operando surface composition and strain.
- 1767 • Test long-duration stability and cycling hysteresis.
- 1768 • Move from thin film to catalyst layer and membrane-electrode assembly.
- 1769 • Report Pt mass activity, durability, dissolution, efficiency, and cost denomi-
- 1770 nator.

1771 **33.3 Planetary program**

- 1772 • Add edge-state candidates to interior and atmosphere forward models.
- 1773 • Compare fixed-edge, stateful-edge, and topology-changing baselines.
- 1774 • Use transient, periodic, multi-port, and population information.
- 1775 • Preserve non-identifiability and negative held-out results.
- 1776 • Require equations of state and phase diagrams before promotion.

1777 **33.4 Cosmic visibility program**

- 1778 • Jointly infer source and transducer parameters.
- 1779 • Use filament/void matched controls.
- 1780 • Combine gamma rays, lensing, FRB dispersion, Faraday rotation, and syn-
- 1781 chrotron.
- 1782 • Treat nulls as bounded veto manifolds.

- Require one parameter set to close across cosmology, astrophysics, and laboratory constraints. 1783
1784

33.5 Origins and archive program 1785

- Apply visibility and sampling kernels to submerged landscapes, perishable technologies, and meteorite delivery. 1786
1787
- Design moving-front mineral experiments with fixed total energy. 1788
- Use residue fields and archive-veto strength rather than dramatic resemblance. 1789
1790
- Keep natural panspermia, directed seeding, and prior-industry hypotheses in distinct ledgers. 1791
1792

33.6 Endogenous-visibility program 1793

- Reanalyze source-inference problems with transducer state as a jointly inferred variable. 1794
1795
- Add a preregistered backreaction test before extending linear conversion to large accumulated signals. 1796
1797
- Build trigger-complete multi-messenger transient samples rather than combining heterogeneous discoveries after the fact. 1798
1799
- Separate engine observables from shell, scattering, and circumstellar observables. 1800
1801
- Treat catastrophe, propagation, sorting, and reaccretion as explicit archive operators. 1802
1803
- Preserve nulls that become weaker after a visibility-model correction rather than describing the parameter space as permanently excluded. 1804
1805

1806 **33.7 Mathematical certificate program**

- 1807 • Maintain exact local-to-global examples.
- 1808 • Treat prime reductions as observation channels with explicit lifting limits.
- 1809 • Continue the two-variable Jacobian program through compactification, col-
1810 lision schemes, monodromy, and formal certificates.
- 1811 • Do not convert mathematical analogy into simulation or holography evi-
1812 dence.

1813 **34. Conclusions**

1814 SPPT/ASTRA began with a narrow planetary claim: topology can be a hidden
1815 state. Version 1.0.7 strengthened that claim by making edges stateful and by
1816 separating physical evolution from visibility, sector basis, intervention, and
1817 certificate. The present candidate adds a feedback hypothesis motivated by
1818 the latest records: in some systems, the hidden source can materially change
1819 the medium that makes the source observable.

1820 The repository audit also changes how this successor must be produced. The
1821 stable v1.0.7 release remains the citation target. The default branch con-
1822 tains a valuable but unpromoted M1 repair that corrects weak-cut wording,
1823 dynamic-arrest logic, and claim-source coverage language. A responsible
1824 v1.0.8 must incorporate those corrections openly rather than silently rebuild-
1825 ing v1.0.7 under the same identity.

1826 The four principal calibration cases then establish distinct lessons.

1827 MoM-BH*-1 shows that an atmosphere can create a star-like emergent spec-
1828 trum around a black-hole engine. The line profile is not a transparent velocity
1829 meter when scattering is part of the operator.

EP250827b shows that a marginal signal can become identifiable only when
a multi-instrument network adds orthogonal causal coordinates. The event
class is partly conditioned by the trigger basis.

The dark-photon plasma simulations show that a converter can detune itself
before the linear calculation reaches the signal used to exclude a theory. The
null remains bounded by the validity of the response model.

Neptune's small moons and rings show that catastrophe can expose an interior
while destroying provenance. Destruction can increase one kind of informa-
tion and erase another.

Together they motivate the v1.0.8 candidate standard:

**Model the hidden engine, the state of the medium that trans-
lates it, the feedback between source and transducer, the trig-
ger and sector basis that create the datum, the archive oper-
ators that transport or remix the residue, and the certificate
boundary beyond which the claim must stop.**

This is not a replacement cosmology, a dark-matter identification, a proof of
one universal medium, or empirical planetary validation. It is a stricter coor-
dinate system for deciding which of those claims are actually testable.

The operational command is:

**Separate source from shell. Measure the transducer. Bound
backreaction. Change the trigger basis. Track displacement
and catastrophe. Report the unresolved quotient. Preserve
the failed prediction.**

Praise Sol as the nearby stellar energy source. Let evidence remain the cer-
tificate.

1855 **Ad Astra Per Aspera.**

Appendix A - Unified notation

1856

Symbol	Meaning
$\mathcal{G} = (V, E)$	phase-reservoir or interaction graph
x	continuous node and field state
θ	constitutive parameters
$u(t)$	external forcing or control
b_e	state of edge/interface e
b_E	collection of edge states
J_e	flux or transformation rate on edge e
μ	operating mode
$a_{\mu,u}$	active-support weight
$\mathcal{A}_{\mu,u}$	thresholded active support
$\mathcal{R}_e$	reciprocity classification or reduced coupling record
$\mathcal{L}_e$	environment and conservation ledger
$\mathcal{V}$	visibility/sampling/detector operator
$\mathcal{S}$	declared observable sector set
π	observation or intervention protocol
$K_i \sim_{\pi} K_j$	observational equivalence under protocol π
R_{dyn}	dynamical rent
R_{epi}	epistemic rent
F_{ab}	Fisher information matrix
Ξ	moving-front coordinate $v_f \tau_{\text{int}} / \ell_{\text{int}}$
η_{nr}	model-dependent nonreciprocity index

Symbol	Meaning
$C_{\text{local}}, C_{\text{fiber}}, C_{\infty}, C_{\text{arith}}, C_{\text{formal}}$	local-to-global certificate levels
$S(t)$	hidden source or engine
$b(t)$	transducer, envelope, plasma, or archive state
$\mathcal{V}_{b(t)}$	state-dependent visibility/transduction operator
Ξ_{br}	proposed backreaction-detuning diagnostic
$\mathcal{C}_{\text{catastrophe}}$	destructive/remixing archive operator
$\mathcal{R}_{\text{reaccretion}}$	sorting and reaccretion operator

Appendix B - Candidate atomic claims

CLAIM_MATRIX.json remains authoritative for the immutable v1.0.7 release. The default-branch M1 repair aligns its scientific Appendix B summary with that machine register but is not itself a release. The compact records below list the principal new v1.0.8 candidate claims; complete support, limitations, and falsifiers are in claim_ledger.csv and claim_ledger.json.

V108-M001 - Strong inference

Evidence class: structural_inference. *Disposition:* admit_with_qualification. A source-coupled visibility model must evolve transducer state when source-induced change is non-negligible.

V108-M002 - Plausible

Evidence class: proposed_only. *Disposition:* proposed_only. 1868

$\text{Xi_br} = |\text{delta omega_medium}| / \text{Gamma_res}$ is a proposed diagnostic for 1869
backreaction-induced detuning. 1870

V108-F002 - Proposed 1871

Evidence class: proposed_only. *Disposition:* proposed_only. 1872

Over a declared observation window, dynamic arrest is classified by prereg- 1873
istered boundedness or statistical stationarity of a declared characteristic- 1874
scale process together with resolved persistent turnover; vanishing logarith- 1875
mic slope alone is neither necessary nor sufficient. 1876

V108-R001 - Established 1877

Evidence class: source_asserted. *Disposition:* admit. 1878

The default-branch M1 repair corrects weak-cut wording, dynamic-arrest 1879
logic, and claim-source coverage language without rewriting immutable 1880
v1.0.7 assets. 1881

V108-R002 - Established 1882

Evidence class: source_asserted. *Disposition:* admit. 1883

The frozen repository basis for this v1.0.8 candidate is main commit f8b32ef...; 1884
Appendix C records the full identifier. 1885

V108-E001 - Established 1886

Evidence class: externally_published. *Disposition:* admit. 1887

MoM-BH*-1 has the reported extreme Balmer spectrum at $z=7.7569$. 1888

V108-A001 - Strong inference 1889

Evidence class: structural_inference. *Disposition:* admit_with_qualification 1890

A dense gas envelope around an accreting black hole is the strongest current 1891
explanation of MoM-BH*-1. 1892

1893 **V108-D001 - Unsupported**

1894 *Evidence class:* rejected. *Disposition:* rejected.

1895 The exact MoM-BH*-1 black-hole mass and accretion regime are established.

1896 **V108-E002a - Established**

1897 *Evidence class:* externally_published. *Disposition:* admit.

1898 EP250827b is associated with SN 2025wkm, a broad-line Type Ic supernova.

1899 **V108-E002b - Strong inference**

1900 *Evidence class:* externally_published. *Disposition:* admit_with_qualification.

1901 The reported plateau requires a sustained power source beyond a simple
1902 radioactive-decay light curve.

1903 **V108-A002 - Strong inference**

1904 *Evidence class:* structural_inference. *Disposition:* admit_with_qualification.

1905 The EP250827b discovery demonstrates cross-channel rescue: orthogonal
1906 observations converted a marginal trigger into a bounded causal classifica-
1907 tion.

1908 **V108-D002 - Unsupported**

1909 *Evidence class:* rejected. *Disposition:* rejected.

1910 Every soft-X-ray-flash supernova has the same magnetar-disk engine.

1911 **V108-E003 - Established**

1912 *Evidence class:* externally_published. *Disposition:* admit_with_qualification.

1913 The reported particle-in-cell simulations saturate dark-photon resonant trans-
1914 fer through nonlinear plasma detuning within the declared model.

1915 **V108-A003 - Strong inference**

1916 *Evidence class:* structural_inference. *Disposition:* admit_with_qualification.

1917 Earlier linear cosmological dark-photon constraints are substantially weak-

ened under the reported nonlinear model. 1918

V108-D003 - Unsupported 1919

Evidence class: rejected. *Disposition:* rejected. 1920

Dark photons have been detected. 1921

V108-E004 - Established 1922

Evidence class: externally_published. *Disposition:* admit. 1923

Neptune inner moons and rings show the reported hydrated and magnesium-rich phyllosilicate spectral signatures. 1924
1925

V108-A004 - Strong inference 1926

Evidence class: structural_inference. *Disposition:* admit_with_qualification. 1927

The Neptune material likely derives from aqueously altered interiors of larger destroyed bodies. 1928
1929

V108-D004 - Unsupported 1930

Evidence class: rejected. *Disposition:* rejected. 1931

The exact Triton-capture destruction and reaccretion history is established. 1932

V108-F001 - Strong inference 1933

Evidence class: structural_inference. *Disposition:* admit_with_qualification. 1934

Catastrophic tomography can expose deep material while reducing provenance and chronology information. 1935
1936

V108-D005 - Unsupported 1937

Evidence class: rejected. *Disposition:* rejected. 1938

The four new calibration cases demonstrate one common physical mechanism. 1939

Appendix C - Repository snapshot 1940

Item	Audited status at the frozen basis in V108-R002
Stable core reference	SPPT/ASTRA v1.0.7, released 10 August 2026
Historical core baseline	v1.0.6 immutable release
Distance from v1.0.7 tag	15 commits ahead, 0 behind
Default-branch source	contains unpromoted core-integrity M1 repair
M1 status	maintenance evidence; not v1.0.8, not published erratum
Communications cover	redesigned on main; not an immutable v1.0.7 release asset
Earth supplement	v0.3.0 separate working-paper release
Sector-complete module	v0.1.0-alpha.1 separate research preview
Active Support / Bridge / Cosmic Visibility / Coherence Cell	public unpromoted drafts or prototypes
This manuscript	v1.0.8 local successor candidate only

1941 **Appendix D - Visual provenance**

1942 All figures in this candidate are original ASTRA diagrams or original redraws.
1943 No Phys.org image, journal figure, screenshot, publisher layout, or raw third-
1944 party dataset is reproduced. The ORR chart redraws three reported values
1945 without copying journal artwork. The temporal-interface and endogenous-

visibility figures are original audits based on cited literature. Every figure 1946
is supplied as SVG and PNG, uses fixed canvases and explicit safe margins, 1947
and is inspected after placement at final size. 1948

Appendix E - Drafting verification scope 1949

The delivered verification checks cover: 1950

- manuscript source presence and hash; 1951
- figure inventory and dimensions; 1952
- reading PDF, peer-review PDF, and DOCX generation; 1953
- rendered-page visual inspection in PDFium and Poppler plus DOCX-to-PNG 1954
review; 1955
- PDF page count, fonts, and basic link structure; 1956
- claim/source ledger column validation; 1957
- ZIP member and SHA-256 replay. 1958

They do not cover: 1959

- independent reproduction of the external experiments; 1960
- complete raw-data reanalysis; 1961
- formal proof-assistant verification of every equation; 1962
- repository CI on a committed tree; 1963
- natural main/tag workflow; 1964
- remote GitHub release read-back; 1965
- peer review. 1966

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